

Received 30 July 2026, accepted 22 August 2026, date of publication 1 September 2026, date of current version 8 September 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3728424

RESEARCH ARTICLE

State-of-Charge Estimation of Lithium-Ion Batteries Using a Computationally Efficient Particle Filter

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This work was supported in part by the National Research Council of Thailand (NRCT), and in part by Mae Fah Luang University under Grant N42A690444.

ABSTRACT Accurate state of charge (SOC) estimation is essential for battery management systems (BMS) in electric vehicles (EVs), especially under nonlinear dynamics, non-Gaussian noise, and highly dynamic operating conditions. Particle filters (PFs) are well suited to these challenges; however, conventional implementations such as the sampling importance resampling particle filter (SIRPF) often suffer from weight degeneracy, particle impoverishment, and high computational cost, which limit their suitability for real-time embedded applications. To overcome these issues, this paper proposes a percentile-based resampling (PBR) strategy integrated into the PF framework, resulting in a percentile-based particle filter (PBPF) that reformulates the resampling mechanism by retaining percentile-ranked informative particles and regulating particle replication according to their importance weights. The proposed PBPF is evaluated using the widely adopted LA92 driving cycle at 40°C and further validated under additional operating temperatures of 25°C and 0°C, as well as the NASA B0005 battery at the 10th discharge cycle. Comparative results against the EKF, AEKF, conventional SIRPF, and extended particle filter (EPF) demonstrate that the proposed PBPF consistently achieves superior SOC and terminal voltage estimation performance. Specifically, the proposed PBPF achieves the best SOC RMSE values of **1.308%**, **0.405%**, and **0.236%** for the LA92 driving cycle at 40°C, 25°C, and 0°C, respectively, and **0.07%** for the NASA B0005 battery at the 10th discharge cycle. These correspond to the highest overall SOC improvements of **26.19%**, **74.79%**, **83.57%**, and **98.66%** over the EKF/dual EKF baseline, respectively. These results demonstrate that the proposed PBPF provides a favorable accuracy–complexity trade-off, making it well suited for real-time embedded BMS applications.

INDEX TERMS State of charge estimation, battery management system, lithium-ion battery, Bayesian filtering, particle impoverishment, percentile-based resampling, Kalman filter, signal processing.

I. INTRODUCTION

Electric vehicles (EVs) rely on battery management systems (BMS) to ensure safe and efficient operation by monitoring key internal states, including the state of charge (SOC), state of health (SOH), and cell temperature [1], [2]. Among these variables, SOC plays a critical role in energy management and driving range prediction; however, it cannot be directly measured and must instead be inferred using model-based

estimation algorithms. Accurate SOC estimation is also essential for broader energy storage applications, including grid-integrated battery systems and renewable energy buffering. Reliable SOC estimation enables optimal energy dispatch, improved safety, and enhanced battery lifetime management, which are crucial for large-scale electrification and sustainable energy systems.

Conventional SOC estimation methods, such as Coulomb counting (CC) and SOC–open-circuit voltage (SOC–OCV) mapping, are computationally efficient but suffer from accumulated sensor noise, sensitivity to initial SOC uncertainty,

The associate editor coordinating the review of this manuscript and approving it for publication was Yongquan Sun¹.

and degraded accuracy under dynamic operating conditions. These shortcomings are particularly pronounced for batteries with nonlinear or flat OCV–SOC characteristics [3], [4]. To improve estimation performance, equivalent circuit models (ECMs) combined with Bayesian filtering methods, including Kalman filter (KF) and particle filter (PF)-based approaches, have been extensively investigated [5], [6], [7], [8], [9]. Despite these advances, battery parameters remain strongly affected by temperature variations, which influence both available capacity and internal resistance. Nevertheless, many existing SOC estimation studies consider only a limited set of discrete temperature conditions, highlighting the need for robust SOC estimation frameworks that are capable of maintaining reliable performance across a wide range of operating scenarios [10].

In recent years, data-driven and deep learning (DL)-based approaches have attracted increasing attention for SOC and voltage estimation in LIBs. These methods aim to capture the complex nonlinear relationships between measurable signals and battery states directly from data. For instance, unsupervised machine learning (ML) approaches, such as clustering combined with principal component analysis (PCA), can estimate SOC and detect battery failure without prior knowledge of the battery state [11]. Recurrent neural networks (RNNs), including long short-term memory (LSTM), gated recurrent unit (GRU), and Bidirectional LSTM (BiLSTM), can accurately estimate SOC using current, temperature, and average voltage, achieving low errors across varying operating conditions [12]. Other DL architectures, including feedforward neural networks (FNNs), temporal convolutional networks (TCNs), and convolutional neural networks (CNNs), have also been explored for SOC estimation, with CNNs particularly effective in processing complex sensing signals, such as ultrasonic waveform data, for SOC monitoring in large-format LIBs [13], [14].

To further improve estimation accuracy and robustness, hybrid frameworks that combine DL models with classical filtering techniques have also been investigated, where neural networks extract nonlinear battery features while Bayesian filters perform recursive state estimation [15], [16]. More recently, attention-based architectures have been introduced to enable DL models to focus on the most informative temporal features, thereby enhancing their capability to capture complex battery dynamics. Furthermore, physics-informed DL approaches have been proposed to incorporate electrochemical knowledge into data-driven models, improving physical consistency and estimation reliability under diverse operating conditions [17]. Emerging research directions also include advanced learning paradigms such as physics-informed neural networks, multi-task learning, few-shot learning, and continual learning for intelligent BMS [18]. In addition, several studies have explored joint estimation frameworks that simultaneously estimate multiple battery states, such as SOC and SOH, using DL architectures to improve overall battery monitoring performance [19].

Representative examples of these recent developments include advances in both model-based and data-driven battery state estimation frameworks. Adaptive redacted extended Kalman filter (REKF) approaches have incorporated fuzzy logic to regulate the forgetting factor, thereby enhancing robustness against measurement noise and varying operating conditions [20]. More recently, the Uncertainty-aware Causal Fusion Network (UCF-Net) has integrated physics-informed electrochemical features with behavioral risk indicators through a hierarchical Bayesian fusion framework to provide an interpretable and uncertainty-aware battery fault diagnosis framework, further extending the capabilities of intelligent battery management systems [21]. In parallel, DL-based methods have integrated metaheuristic optimization algorithms, such as manta ray foraging optimization and simulated annealing, to automatically optimize LSTM hyperparameters while improving robustness through noise-enhanced training [22].

Despite these advantages, DL-based approaches typically require large training datasets and significant computational resources. Moreover, model retraining can be necessary when operating conditions, battery chemistries, or aging characteristics change, which can limit their practicality for real-time embedded BMS. Consequently, model-based Bayesian filtering methods, such as KFs and PFs, remain attractive for SOC estimation due to their computational efficiency, robustness, and suitability for real-time implementation [23].

PFs are particularly attractive for SOC estimation due to their ability to handle nonlinear dynamics and non-Gaussian noise [24]. In addition to battery state estimation, particle filtering has been widely applied to sequential estimation problems in various fields, including target tracking, robotics, autonomous navigation, signal processing, ocean acoustics, and communication systems [25]. Recent studies have further demonstrated the effectiveness of Rao–Blackwellized particle filters (RBPFs) in communication engineering for complex signal inference problems involving hidden stochastic states and nonlinear dynamics. For example, an extended RBPF framework was introduced for analog variance noise modulation, where hidden signal states and time-varying noise states were jointly estimated to achieve accurate signal reconstruction under stochastic noise conditions [26].

Nevertheless, conventional PFs suffer from several practical limitations when applied to battery SOC estimation. In particular, particle importance weights tend to collapse over time, leading to weight degeneracy, where only a small subset of particles contributes to the posterior distribution. Subsequent resampling operations can further lead to particle impoverishment, where particle diversity is significantly reduced due to repeated duplication of high-weight particles. Additionally, achieving reliable estimation accuracy often requires a large number of particles, increasing computational burden and limiting real-time deployment in embedded BMS.

To mitigate these issues, most existing PF-based SOC estimators primarily focus on improving the proposal density [8], such as extended PF (EPF), unscented PF (UPF), or cubature PF (CPF), in which KF-based methods are embedded to enhance the importance sampling stage of the standard SIRPF [27]. While these advanced PF variants can improve estimation accuracy, they typically increase computational complexity and remain reliant on global resampling, which can lead to particle impoverishment. In this paper, a different perspective is adopted by directly addressing the resampling mechanism itself. A percentile-based resampling (PBR) scheme [28] is incorporated into the PF framework, resulting in a percentile-based particle filter (PBPF) for SOC estimation of LIBs.

The main contributions of this work are summarized as follows:

- 1) A PBPF is developed for SOC estimation of LIBs, where particle replication is regulated using percentile-ranked importance weights rather than conventional global resampling used in conventional SIRPF.
- 2) The proposed PBR scheme reduces particle degeneracy and particle impoverishment while maintaining the dominant posterior distribution characteristics.
- 3) The proposed PBPF maintains particle diversity while reducing computational complexity, improving real-time applicability for embedded BMS.
- 4) Extensive experimental evaluations demonstrate the favorable accuracy–complexity trade-off of the proposed PBPF.

For evaluation, the PBPF is compared against established nonlinear estimators, including the extended Kalman filter (EKF), adaptive EKF (AEKF), conventional SIRPF, and EPF. The proposed method is evaluated using the widely adopted LA92 driving cycle at 40°C, with additional validation at 25°C and 0°C, together with the NASA B0005 cell under the 10th discharge cycle. Other available PF variants, such as the UPF and CPF, are not implemented because the primary objective of this work is to investigate the effectiveness of the proposed PBR strategy while maintaining low computational complexity and real-time feasibility for embedded BMS. Accordingly, SIRPF is selected as the primary PF baseline, whereas EPF is included as an advanced importance-density-based PF benchmark. Comparative evaluations demonstrate that the proposed PBPF achieves a favorable accuracy–complexity trade-off, making it well suited for real-time embedded BMS applications.

The remainder of this paper is organized as follows. Section II presents the battery modeling framework and formulates the SOC estimation problem. Section III introduces the proposed PBR scheme within the PF framework for SOC estimation and details its algorithmic implementation. Section IV describes the experimental setup, including benchmark datasets and evaluation metrics. Section V presents and discusses the comparative results. Finally,

Section VI concludes the paper and outlines directions for future work.

II. BATTERY MODELING AND SOC ESTIMATION PROBLEM FORMULATION

The 2RC ECM in Fig. 1 is widely adopted for SOC estimation due to its favorable trade-off between modeling accuracy and computational complexity. After model parameter identification, the battery dynamics are formulated in discrete-time state-space form, which can be directly embedded into Bayesian filtering frameworks for online SOC estimation using current I , voltage V , and temperature T measurements [10].

A. SOC DEFINITION AND BATTERY EQUIVALENT CIRCUIT MODEL

The SOC indicates the remaining available battery capacity and is defined as the ratio of the remaining capacity (Q_{rem}) to the maximum available capacity (Q_{max}) [2], [5]:

$$SOC(t) = \frac{Q_{rem}(t)}{Q_{max}(t)} \times 100\%. \quad (1)$$

In this work, the true SOC is used solely for performance evaluation and is obtained from the accumulated battery charge. Depending on the experimental dataset, the accumulated charge is either provided directly or reconstructed from the measured current using the CC principle. The true SOC is not supplied to the estimation algorithms.

Lithium-ion batteries exhibit nonlinear electrical behavior due to the underlying electrochemical reactions during charge and discharge. To enable real-time SOC estimation, equivalent circuit models (ECMs) are widely adopted because they provide a favorable trade-off between modeling accuracy and computational complexity [2], [5], [10]. Among various ECMs, the second-order RC (2RC) model is employed in this work because it effectively captures the instantaneous ohmic voltage drop together with the short- and long-term polarization dynamics of the battery.

In Fig. 1, V_{OCV} denotes the open-circuit voltage, V_{term} represents the terminal voltage, R_0 is the internal resistance, and V_1 and V_2 denote the voltages across the two RC networks. Defining the state vector as $\mathbf{x}_k = [SOC_k, V_{1,k}, V_{2,k}]^T$, the discrete-time state-space dynamics of the battery model can be expressed as:

$$SOC(k+1) = SOC(k) - \frac{\eta \Delta t I(k)}{C_n} \quad (2)$$

$$V_{term}(k) = V_{OCV}(k) - V_1(k) - V_2(k) - I(k)R_0 \quad (3)$$

$$V_1(k+1) = e^{-\frac{\Delta t}{R_1 C_1}} V_1(k) + R_1 \left(1 - e^{-\frac{\Delta t}{R_1 C_1}}\right) I(k) \quad (4)$$

$$V_2(k+1) = e^{-\frac{\Delta t}{R_2 C_2}} V_2(k) + R_2 \left(1 - e^{-\frac{\Delta t}{R_2 C_2}}\right) I(k) \quad (5)$$

where $I(k)$ denotes the input current at time step k , Δt is the sampling interval, C_n is the nominal battery capacity, η is the Coulombic efficiency, and R_1 , C_1 , R_2 , and C_2 are the parameters of the two RC networks [5], [10].

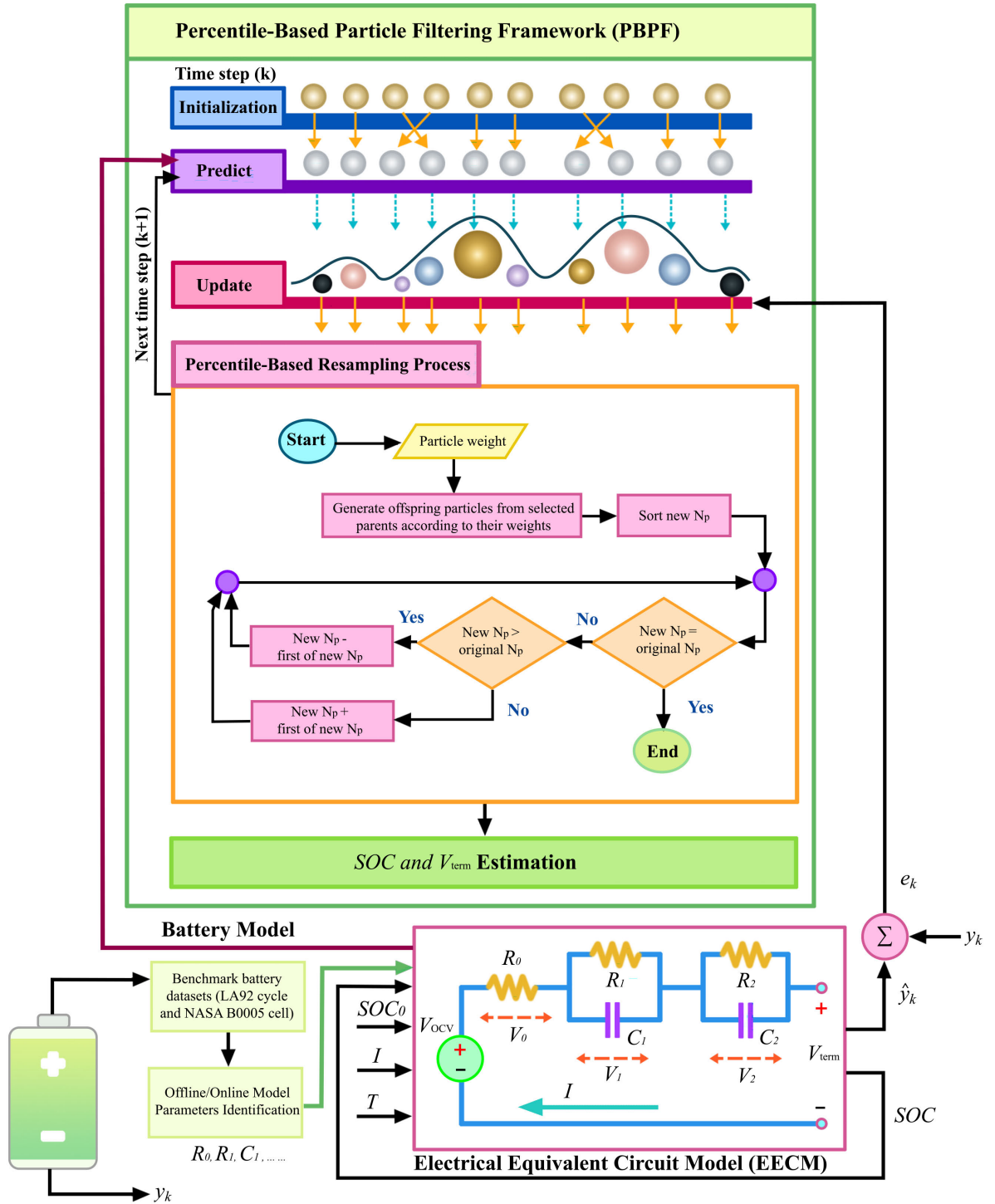


FIGURE 1. Overall structure of the battery model-based SOC estimation framework combining an ECM with a PBPf, including a detailed illustration of the PBR process for $N_p = 10$, where different colors represent the diversity of individual particles.

B. STATE-SPACE REPRESENTATION

The nonlinear state-space representation of the battery model is given by:

$$\mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, u_k) + \mathbf{w}_k, \quad (6)$$

$$y_k = g_k(\mathbf{x}_k, u_k) + v_k, \quad (7)$$

where the state vector is $\mathbf{x}_k = [\text{SOC}_k, V_{1,k}, V_{2,k}]^T \in \mathbb{R}^3$, $u_k = I(k) \in \mathbb{R}$ denotes the input terminal current, and $y_k = V_{\text{term}}(k) \in \mathbb{R}$ represents the measured terminal voltage. The functions $\mathbf{f}_k(\cdot) \in \mathbb{R}^3$ and $g_k(\cdot) \in \mathbb{R}$ represent the nonlinear state transition and measurement functions derived from the SOC dynamics and battery voltage equations. The terms $\mathbf{w}_k \in \mathbb{R}^3$

and $v_k \in \mathbb{R}$ denote the process and measurement noise, respectively, which are assumed to be additive, zero-mean, and mutually independent. For EKF-based methods, these noises are further assumed to be Gaussian with known covariance matrices, consistent with the Kalman filtering framework [27], [29].

For EKF-based methods (EKF and AEKF), the nonlinear battery model is locally linearized around the current state estimate [30]. The resulting Jacobian matrices are used for state propagation and measurement update. As a consequence of this local linearization, EKF-based methods employ the following discrete-time linear state-space representation:

$$\mathbf{x}_{k+1} = \mathbf{A}_k \mathbf{x}_k + \mathbf{B}_k u_k + \mathbf{w}_k, \quad (8)$$

$$y_k = \mathbf{C}_k \mathbf{x}_k + \mathbf{D}_k u_k + v_k, \quad (9)$$

where the system matrices are given by:

$$\mathbf{A}_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-\Delta t/(R_1 C_1)} & 0 \\ 0 & 0 & e^{-\Delta t/(R_2 C_2)} \end{bmatrix}, \quad (10)$$

$$\mathbf{B}_k = \begin{bmatrix} -\frac{\eta \Delta t}{C} \\ R_1 (1 - e^{-\Delta t/(R_1 C_1)}) \\ R_2 (1 - e^{-\Delta t/(R_2 C_2)}) \end{bmatrix}, \quad (11)$$

$$\mathbf{C}_k = \begin{bmatrix} \frac{\partial V_{OCV}}{\partial SOC} & -1 & -1 \end{bmatrix}, \quad \mathbf{D}_k = -R_0. \quad (12)$$

The matrices $\mathbf{A}_k$, $\mathbf{B}_k$, $\mathbf{C}_k$, and $\mathbf{D}_k$ are obtained from the Jacobian matrices of $\mathbf{f}_k(\cdot)$ and $\mathbf{g}_k(\cdot)$ evaluated at the current state estimate [5], [10]. In contrast, PF-based methods propagate particles using the original nonlinear state-space model in Eqs. (6)–(7) without requiring local linearization.

C. CONVENTIONAL PARTICLE FILTER FOR SOC ESTIMATION

The PF approximates the posterior probability density function (PDF) of the system state at time k by representing it with a finite set of weighted samples, commonly referred to as particles. Each particle consists of a state realization $\mathbf{x}_k^i$ and an associated weight w_k^i , forming the set $\{\mathbf{x}_k^i, w_k^i\}_{i=1}^{N_p}$, where N_p is the total number of particles [28]. For the nonlinear state-space model described in Eqs. (6)–(7), the posterior distribution conditioned on the measurement sequence $\mathbf{Y}_k = \{y_1, y_2, \dots, y_k\}$ can be approximated by a discrete weighted sum of Dirac delta functions, expressed as:

$$p(\mathbf{x}_k | \mathbf{Y}_k) \approx \sum_{i=1}^{N_p} w_k^i \delta(\mathbf{x}_k - \mathbf{x}_k^i), \quad (13)$$

where w_k^i denotes the normalized importance weight of the i th particle, with the normalization constraint $\sum_{i=1}^{N_p} w_k^i = 1$.

1) STATE PROPAGATION (PREDICTION STEP)

Assuming a first-order Markov process, each particle is propagated through the nonlinear state transition model in Eq. (6) according to

$$\mathbf{x}_k^i = \mathbf{f}_{k-1}(\mathbf{x}_{k-1}^i, u_{k-1}) + \mathbf{w}_{k-1}^i, \quad (14)$$

which yields a particle-based approximation of the predictive distribution,

$$p(\mathbf{x}_k | \mathbf{Y}_{k-1}) \approx \sum_{i=1}^{N_p} w_{k-1}^i p(\mathbf{x}_k | \mathbf{x}_{k-1}^i), \quad (15)$$

which follows directly from the Chapman–Kolmogorov equation.

2) MEASUREMENT UPDATE (IMPORTANCE WEIGHTING)

Using the measurement model in Eq. (7), the particle weights are updated according to Bayes' rule. In general, the importance weight of each particle is defined as the ratio between the posterior and the proposal distribution. When the prior is chosen as the proposal, this update simplifies to [27]:

$$w_k^i \propto w_{k-1}^i p(y_k | \mathbf{x}_k^i), \quad (16)$$

where $p(y_k | \mathbf{x}_k^i)$ denotes the likelihood of the current measurement given the i th particle.

Assuming additive Gaussian measurement noise, the likelihood function is modeled as:

$$p(y_k | \mathbf{x}_k^i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_k - \hat{y}_k^i)^2}{2\sigma^2}\right), \quad (17)$$

where y_k and $\hat{y}_k^i$ denote the measured V_{term} and the corresponding estimated value obtained from the measurement model for the i th particle, respectively. The parameter σ is the standard deviation of the measurement noise, and $\mathbf{R} = \sigma^2 \in \mathbb{R}^{1 \times 1}$ denotes the corresponding measurement noise covariance matrix [7], [31].

The likelihood function quantifies the consistency between the predicted measurement of each particle and the actual observation, which determines the importance weight assigned to each particle. Particles with predicted measurements closer to the actual observation obtain higher likelihood values and consequently receive larger weights, whereas particles with larger measurement errors receive smaller weights. The particle weights are then normalized as:

$$\hat{w}_k^i = \frac{w_k^i}{\sum_{j=1}^{N_p} w_k^j}, \quad (18)$$

where $\hat{w}_k^i$ denotes the normalized importance weight satisfying $\sum_{i=1}^{N_p} \hat{w}_k^i = 1$. The normalized weights are subsequently used in the resampling process [32].

3) DEGENERACY AND RESAMPLING

The particle filtering procedure described above corresponds to sequential importance sampling (SIS). As the filtering process evolves, the variance of particle weights typically increases, leading to severe weight degeneracy [33], where only a small number of particles retain significant weights. To address this issue, a resampling scheme is commonly employed, resulting in the SIRPF [32], [34], [35].

Resampling is usually activated when the effective sample size (ESS), which provides a quantitative measure of weight

degeneracy, falls below a predefined threshold. The ESS is defined as:

$$N^{\text{eff}} = \frac{1}{\sum_{i=1}^{N_p} (\hat{w}_k^i)^2}, \quad (19)$$

where $\hat{w}_k^i$ denotes the normalized particle weight obtained from the measurement update step.

Although resampling effectively mitigates weight degeneracy, it introduces additional challenges. In particular, repeated replication of high-weight particles and the removal of low-weight ones reduce particle diversity, leading to sample impoverishment [28]. These limitations motivate the PBR strategy proposed in the following section. A comprehensive discussion of PF convergence properties and computational complexity can be found in [36].

III. PROPOSED PERCENTILE-BASED PARTICLE FILTER FOR SOC ESTIMATION

This section introduces a PBR scheme [28] that enhances the sequential Bayesian filtering framework by reducing computational complexity while maintaining efficient performance. The proposed method can be regarded as an adaptive resampling strategy that reduces the computational burden of the standard SIR scheme while preserving, and in some cases improving, estimation performance. PBR selects parent particles by retaining only the most informative particles, as determined by their importance weights computed during the PF update step. In this work, parent particles are selected by retaining the highest-weight particles whose cumulative importance weight satisfies a specified percentile threshold [37]. This selected subset contains the particles that dominate the posterior state probability distribution.

Let $\{w_k^i\}_{i=1}^{N_p}$ denote the set of particle weights at time step k , where N_p is the number of particles and the weights satisfy $\sum_{i=1}^{N_p} w_k^i = 1$. Assume that the weights are arranged in descending order, and define the weight vector as $\mathbf{W}_k = [w_{k,1} \ w_{k,2} \ \dots \ w_{k,N_p}]$, which satisfies $w_{k,1} > w_{k,2} > w_{k,3} > \dots > w_{k,N_p} \geq 0$. In the PBPF formulation, perc is defined as the target cumulative percentage of particle weights retained during the particle selection stage. The value of perc determines the number of high-weight particles selected for offspring generation. To avoid confusion, note that perc refers to the target cumulative weight, while $\mathbf{W}_{k,\text{perc}}$ denotes the set of selected particle weights. This criterion is satisfied under the following conditions:

$$\text{perc} \geq \sum_{j=1}^{N_{\text{perc}}} w_{k,j}, \quad (20)$$

where N_{perc} denotes the number of highest-weight particles whose cumulative importance weight satisfies the percentile threshold perc. In practice, N_{perc} is substantially smaller than N_p . In the limiting case where $N_{\text{perc}} = 1$, the posterior distribution collapses to a single dominant particle, which is

then selected as the sole parent for generating all offspring particles. When $N_{\text{perc}} > 1$, the selected particles are treated as a set of parent particles from which offspring particles are generated. The number of offspring assigned to each parent is determined by its normalized importance weight. Specifically, a particle with a larger weight (e.g., $w_{k,1} > w_{k,2}$) generates more offspring; consequently, particle $\mathbf{x}_{k,1}$ is duplicated more frequently than particle $\mathbf{x}_{k,2}$. After offspring generation, the total number of particles remains unchanged.

Let $\mathbf{W}_{k,\text{perc}} = \{w_{k,1}, w_{k,2}, \dots, w_{k,N_{\text{perc}}}\}$ denote the set of weights corresponding to the particles selected based on the percentile threshold perc. Each parent particle constructs a corresponding number of offspring particles, denoted by $n_1, n_2, \dots, n_{N_{\text{perc}}}$, such that $\sum_{i=1}^{N_{\text{perc}}} n_i = N_p$. Intuitively, the number of offspring particles generated from each selected parent particle is assigned in proportion to its corresponding weight, i.e.,

$$n_i = \left\lceil \frac{w_{k,i}}{\sum_{j=1}^{N_{\text{perc}}} w_{k,j}} N_p \right\rceil, \quad (21)$$

where $\lceil \cdot \rceil$ denotes the ceiling operator, which rounds its argument up to the nearest integer.

In the proposed scheme, offspring generation is governed by the importance weights of the parent particles. Specifically, parent particles with larger weights are allowed to generate more descendants, while those associated with smaller weights yield fewer offspring. It is worth noting that, in practice, the total number of generated particles $\sum_{j=1}^{N_{\text{perc}}} n_j$ does not necessarily equal N_p , which can cause the total number of particles to vary across iterations. To preserve a fixed number of particles N_p , excess offspring particles are discarded, or additional particles are replicated when required. An illustration of the particle alteration mechanism employed in the PBR framework is provided in Fig. 1. Following this adjustment, the PBR yields a refined set of particles that are primarily derived from high-quality parent particles. These particles are then carried forward to the next iteration of the PF. At each time step, the state estimate, including the SOC and terminal voltage, is obtained from the resampled particle set as the weighted average of the corresponding particle states.

The percentile threshold is specified as a user-defined parameter in the proposed PBPF. The influence of the percentile threshold on estimation performance is investigated through the sensitivity analysis presented in Section V-A3. Prior research [28], [37] suggests that selecting particles according to a cumulative-weight percentile threshold effectively captures the dominant posterior distribution while reducing computational burden. Future work could explore adaptive percentile selection and quantitative evaluation of particle diversity (e.g., ESS or particle entropy) to further improve SOC estimation under diverse operating conditions. The experimental setup for validating the proposed PBPF is presented in the following section.

IV. EXPERIMENTAL SETUP

The performance of the proposed SOC estimation methods is evaluated using two publicly available LIB datasets commonly adopted in the literature: (i) LA92 driving cycle at 40 °C from the Turnigy Graphene 5000 mAh 65 C cell, representing highly dynamic automotive load conditions [10], [38], [39], and (ii) NASA B0005 cell using the 10th discharge cycle, capturing realistic constant-current discharge behavior [40], [41]. These datasets enable evaluation of the robustness of the proposed algorithm under distinct operating profiles and dataset characteristics. In addition to the conventional EKF, AEKF, and SIRPF methods, the EPF is included as an advanced particle filtering benchmark to provide a more comprehensive comparison of estimation accuracy and computational efficiency. [8].

For the proposed PBPF, the cumulative-weight percentile threshold was set to $\text{perc} = 90\%$ for the LA92 dataset at 40 °C and the NASA B0005 dataset, while a threshold of $\text{perc} = 92\%$ was adopted for the LA92 dataset at 25 °C and 0 °C using the same particle selection strategy [28]. Unlike the LA92 dataset, which is evaluated under multiple operating temperatures (40 °C, 25 °C, and 0 °C), the NASA B0005 dataset is available only at a fixed operating temperature of approximately 24 °C.

The LA92 and NASA B0005 datasets correspond to different lithium-ion cells with distinct capacities, chemistries, and operating conditions. Specifically, the LA92 dataset uses a Turnigy Graphene 5000 mAh 65 C cell under dynamic automotive load conditions, whereas the NASA B0005 dataset uses a different commercial cell with a nominal capacity of 2.0 Ah. For the selected constant-current discharge cycle, an initial capacity of 1.856 Ah is used in the estimation model. While the datasets differ in brand and parameters, this variation is intentional to evaluate the robustness of the proposed PBPF method across diverse operating profiles. All filters are implemented using dataset-specific battery models and parameters to ensure that the observed performance differences arise from the filtering algorithms rather than artificially matched battery characteristics.

TABLE 1. EKF-based parameter settings for the LA92 driving cycle at 40 °C.

Parameter	Symbol	Value / Description
Sampling interval	Δt	1 s; downsampled from 0.1 s
Nominal capacity	Q_n	4.81 Ah; fixed
Initial SOC	SOC_0	1.0 (100%)
Initial covariance	P_0	$\text{diag}(0.025, 0.01, 0.01)$
Process noise	Q	$\text{diag}(10^{-6}, 10^{-5}, 10^{-5})$
Measurement noise	R	2.5×10^{-5}
SOC-OCV model	–	11th-order polynomial
ECM	–	2RC ECM with known parameters

A. LA92 DRIVING CYCLE: DYNAMIC CURRENT EXCITATION

The LA92 driving current profile is used to evaluate SOC estimation performance under highly dynamic automotive load conditions. The dataset provides measured terminal voltage V_{term} , current I , temperature T , and accumulated charge Q (in Ah). In this study, the current, terminal voltage, and temperature are used as filter inputs, while the true SOC

TABLE 2. EKF-based parameter settings for the NASA B0005 cell (Discharge Cycle 10).

Parameter	Symbol	Value / Description
Sampling interval	Δt	Nonuniform; median value used
Nominal capacity	Q_n	2.0 Ah; fixed
Initial capacity estimate	$Q_{n,0}$	1.856 Ah; estimated online
Temperature	T	24 °C
Initial SOC	SOC_0	1.0 (100%)
Initial covariance	$P_{x,0}$	$\text{diag}(0.01, 0.001, 0.001)$
State process noise	Q_x	$\text{diag}(10^{-5}, 10^{-6}, 10^{-6})$
State measurement noise	R_x	2.0×10^{-5}
Estimated parameters	θ_k	$[R_0 \ Q_n]^T$
Parameter process noise	Q_θ	$\text{diag}(10^{-7}, 10^{-8})$
Parameter measurement noise	R_θ	1.0×10^{-2}
ECM	–	2RC ECM with estimated parameters
Estimator	–	Dual EKF

is computed offline from the accumulated charge according to Eq. (1). All signals are downsampled from 0.1 s to 1 s prior to estimation to match the discrete-time filter implementation.

A 2RC ECM is adopted, with SOC and two polarization voltages: V_1 and V_2 , defined as the state vector, $\mathbf{x}_k = [SOC_k \ V_{1,k} \ V_{2,k}]^T$. Model parameters are obtained through offline interpolation as a function of SOC (i.e., $V_{OCV} = f(SOC)$) using dataset-provided characterization data at 40 °C, and are kept fixed during the estimation process. The OCV is represented using an 11th-order polynomial fitted to the SOC-OCV data. All experiments in this study use this fixed 40 °C characterization; the effects of temperature variation on estimation performance are not considered in this work.

An EKF is employed as the baseline estimator with the initial SOC set to 100%. The initial state covariance P_0 , process noise covariance Q , and measurement noise covariance R matrices are manually tuned to ensure stable convergence under dynamic current excitation. These settings are shared by all EKF-based methods (EKF and AEKF), while EPF employs the same battery model and initialization but uses the EKF to construct the importance density for particle propagation. The corresponding model and filter initialization settings are summarized in Table 1.

For PF-based methods (SIRPF, EPF, and PBPF), a common set of Q and R is used across all filters. These covariances are tuned independently from the EKF-based methods to account for the different uncertainty representations inherent in particle filtering frameworks. The EPF employs the EKF-based importance density to improve particle propagation, whereas the proposed PBPF adopts the proposed PBR strategy. All methods employ the same battery model, initialization, and experimental data, differing only in their respective filtering mechanisms.

B. NASA B0005 CELL: CONSTANT-CURRENT DISCHARGE

Experiments are further conducted using the NASA B0005 cell, with 10th discharge cycle selected to represent a realistic constant-current discharge scenario. The dataset provides measured V_{term} and I sampled at nonuniform time intervals. All invalid samples are removed, and only

the discharge segment ($I < 0$) is retained for analysis. A representative sampling interval is obtained using the median of recorded time difference and is used for discrete-time filter implementation. The measured I and V_{term} are used as filter inputs in our experiments. Since the NASA B0005 dataset does not provide a true SOC directly, the true SOC is reconstructed offline using the CC method based on the measured current and nominal battery capacity. The resulting true SOC is used solely for performance evaluation.

The same 2RC ECM is adopted, with SOC and two polarization voltages defined as the state vector $\mathbf{x}_k$. For the NASA dataset, the OCV is modeled as a static nonlinear function of SOC, i.e., $V_{\text{OCV}} = f(\text{SOC})$, represented by a polynomial fitted to SOC–OCV characterization data. Unlike the LA92 case, a dual EKF (DEKF) structure is employed to jointly estimate SOC and selected battery parameters, including internal resistance and capacity to account for increased parameter uncertainty and the absence of dedicated characterization data.

For fair comparison, the same battery model, parameter-estimation structure, and initialization settings are retained for all methods, including DEKF, AEKF-EKF, SIRPF-EKF, EPF-EKF, and the proposed PBPF-EKF approaches. Across these methods, the parameter-estimation EKF remains unchanged, while different filtering strategies are applied only to the SOC and V_{term} estimation stage. In particular, EPF-EKF employs an EKF-based importance density for particle propagation, whereas PBPF-EKF employs the proposed PBR strategy. The $\mathbf{Q}$ and $\mathbf{R}$ of the state estimation filters are identical within each filter family (EKF/AEKF and SIRPF/EPF/PBPF) but tuned separately between families. The corresponding settings are provided in Table 2.

C. PERFORMANCE EVALUATION METRICS

In this section, the performance of the proposed method is evaluated in comparison with three other filters using root mean square error (RMSE), mean absolute error (MAE), and maximum error (MaxE) metrics for both SOC and V_{term} estimation. The MAE reflects estimation accuracy, RMSE indicates effectiveness and robustness, and MaxE captures the worst-case deviation at each time instant. The true SOC is used as the ground truth for error computation. The corresponding mathematical formulations of these performance indices are given as follows [39]:

$$\text{RMSE} = \begin{cases} \sqrt{\frac{1}{K} \sum_{k=1}^K (\text{SOC}_k - \widehat{\text{SOC}}_k)^2} \\ \sqrt{\frac{1}{K} \sum_{k=1}^K (V_{\text{term},k} - \widehat{V}_{\text{term},k})^2} \end{cases} \quad (22)$$

$$\text{MAE} = \begin{cases} \frac{1}{K} \sum_{k=1}^K |\text{SOC}_k - \widehat{\text{SOC}}_k| \\ \frac{1}{K} \sum_{k=1}^K |V_{\text{term},k} - \widehat{V}_{\text{term},k}| \end{cases} \quad (23)$$

and

$$\text{MaxE} = \begin{cases} \max_k |\text{SOC}_k - \widehat{\text{SOC}}_k| \\ \max_k |V_{\text{term},k} - \widehat{V}_{\text{term},k}| \end{cases} \quad (24)$$

where K denotes the total number of data samples, SOC_k and $\widehat{\text{SOC}}_k$ represent the actual and estimated SOC at sampling time k , respectively, while $V_{\text{term},k}$ and $\widehat{V}_{\text{term},k}$ denote the actual and estimated V_{term} at sampling time k . The following section presents the experimental results and discusses the performance of the proposed PBPF through comprehensive comparisons with the benchmark methods.

V. RESULTS AND DISCUSSIONS

In this section, the performance of the proposed PBPF method is compared with other traditional SOC estimators, including EKF, AEKF, SIRPF, and EPF, using the LA92 driving cycle at 40 °C and the NASA B0005 cell at the 10th discharge cycle. Additional validation is conducted using the LA92 driving cycle at 25 °C and 0 °C. The comparison is conducted across different numbers of particles for PF-based methods (SIRPF, EPF, and PBPF), with $N_p \in \{200, 500, 1000, 2000\}$. The evaluation focuses on the accuracy and robustness of SOC and V_{term} estimation based on the error metrics defined in the previous section.

A. RESULTS ON LA92 DRIVING CYCLE AT 40 °C

The LA92 driving cycle is a highly dynamic load profile widely used to evaluate battery state estimation algorithms under realistic conditions. It contains approximately 149,610 samples with a sampling interval of 1 s, corresponding to about 4.5 h of operation, and exhibits frequent current variations that pose significant challenges for accurate SOC and V_{term} estimation. Reliable tracking of SOC and V_{term} under such conditions is important for BMS to support safe, reliable, and efficient operation.

For the LA92 case, all evaluations are performed at a fixed temperature of 40 °C to ensure a fair comparison among estimation methods and consistent with the available experimental data. A single filter is employed to simultaneously estimate both SOC and V_{term} using the same measurement set. The estimation performance is illustrated using four figures: SOC estimation (Fig. 2), SOC error (Fig. 3), V_{term} estimation (Fig. 4), and V_{term} error (Fig. 5). In these figures, the true SOC or the measured V_{term} from the dataset is shown by a black solid line, while EKF, AEKF, SIRPF, EPF, and PBPF results are depicted in light red, green, dark blue, light blue, and magenta, respectively.

For PF-based methods, each figure compares SOC and V_{term} estimation or their corresponding estimation errors under different particle numbers: (a) $N_p = 200$, (b) $N_p = 500$, (c) $N_p = 1000$, and (d) $N_p = 2000$.

1) SOC TRACKING PERFORMANCE

All methods are evaluated under identical experimental conditions. Within each filter family, identical noise covariance

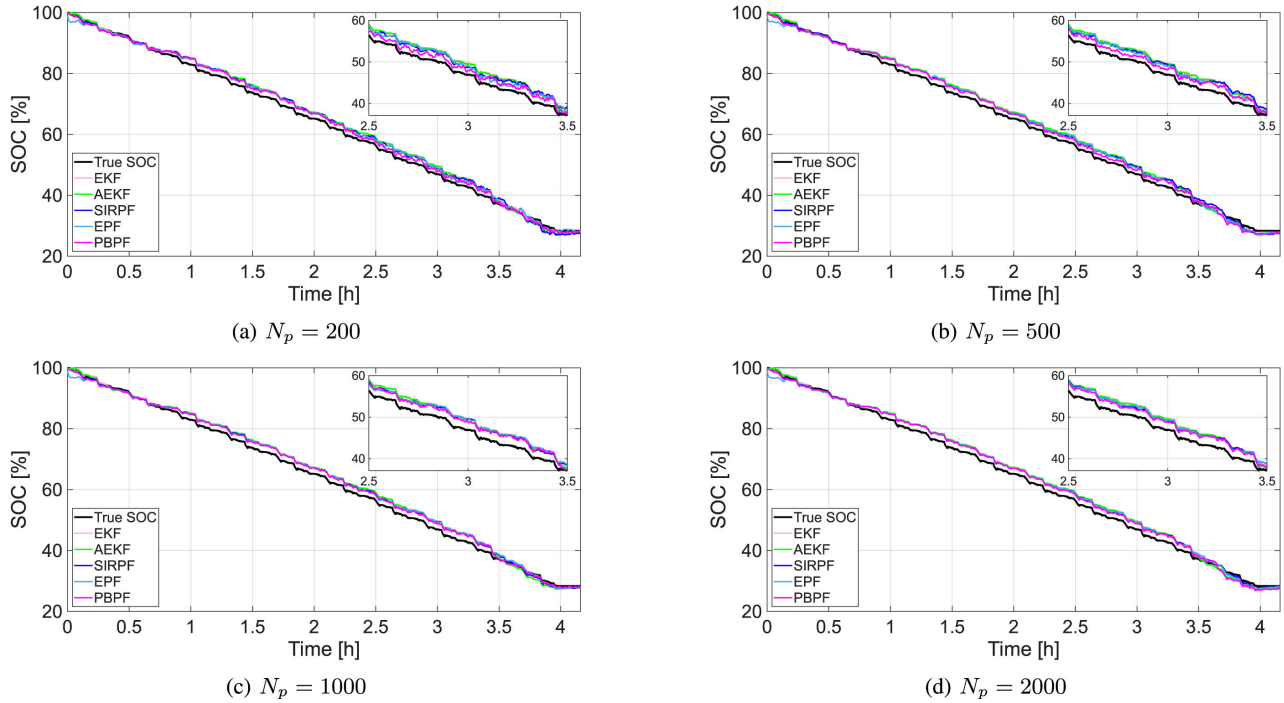


FIGURE 2. SOC estimation with varying particle numbers for the LA92 driving cycle at 40 °C.

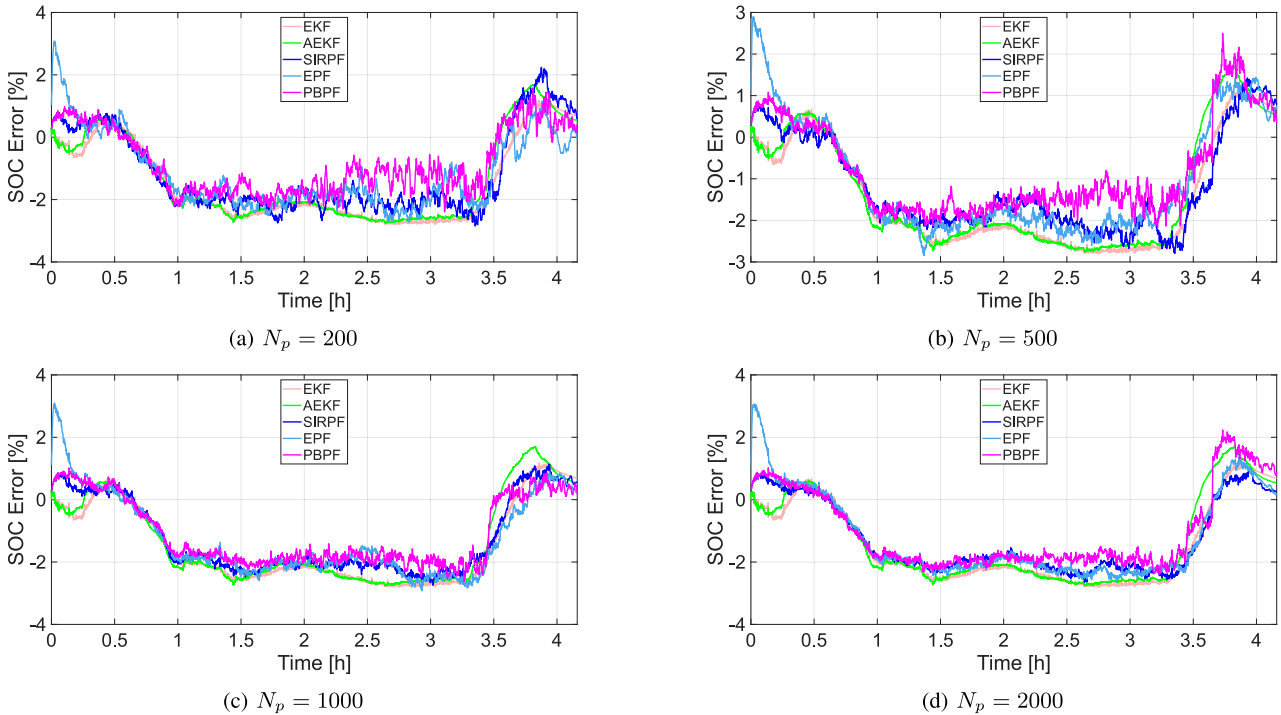


FIGURE 3. SOC estimation error with varying particle numbers for the LA92 driving cycle at 40 °C.

settings are used to ensure that observed performance differences arise from the filtering strategies rather than tuning advantages.

Fig. 2 illustrates the SOC estimation results over time (in h) for all methods. In case (a), the proposed PBPF achieves accurate SOC tracking with a relatively small number of

particles, closely following the true SOC. A zoomed-in view between 2.5 and 3.5 h highlights that the PBPF provides the closest match to the reference trajectory, capturing fine SOC variations more effectively than the other methods. Compared with the conventional SIRPF, the EPF exhibits improved SOC tracking by employing an EKF-based importance

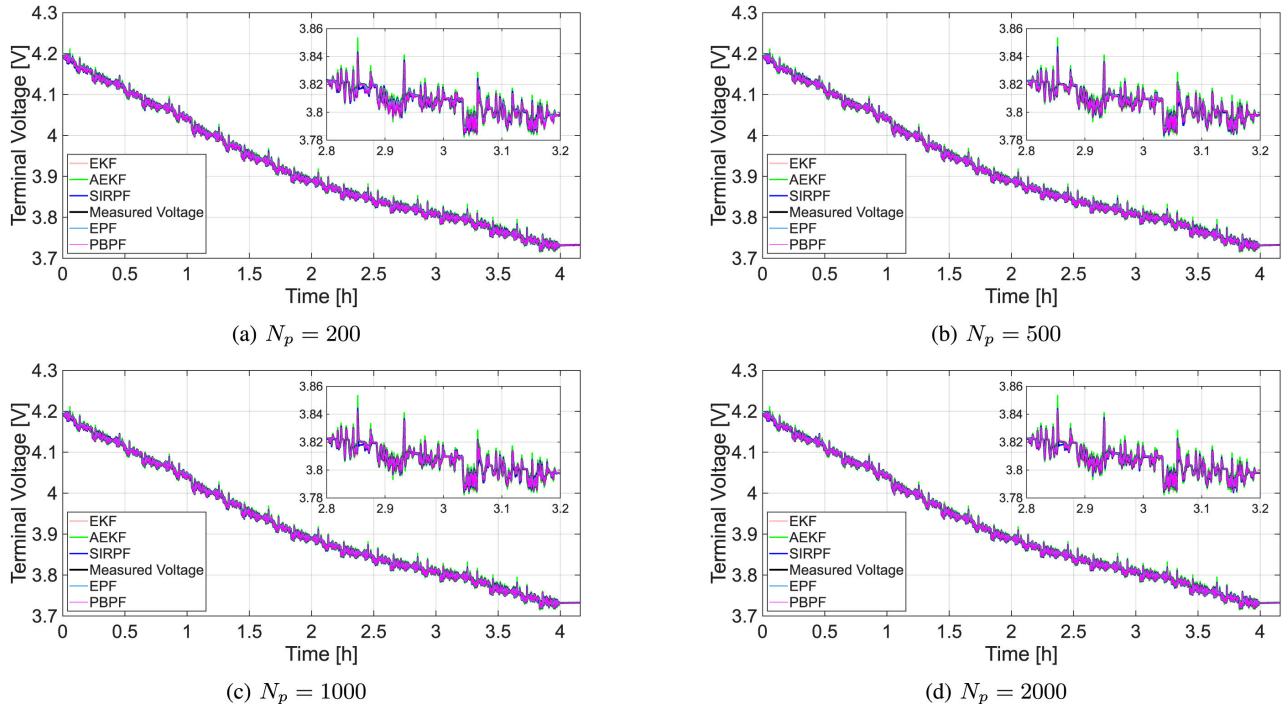


FIGURE 4. Terminal voltage estimation with varying particle numbers for the LA92 driving cycle at 40 °C.

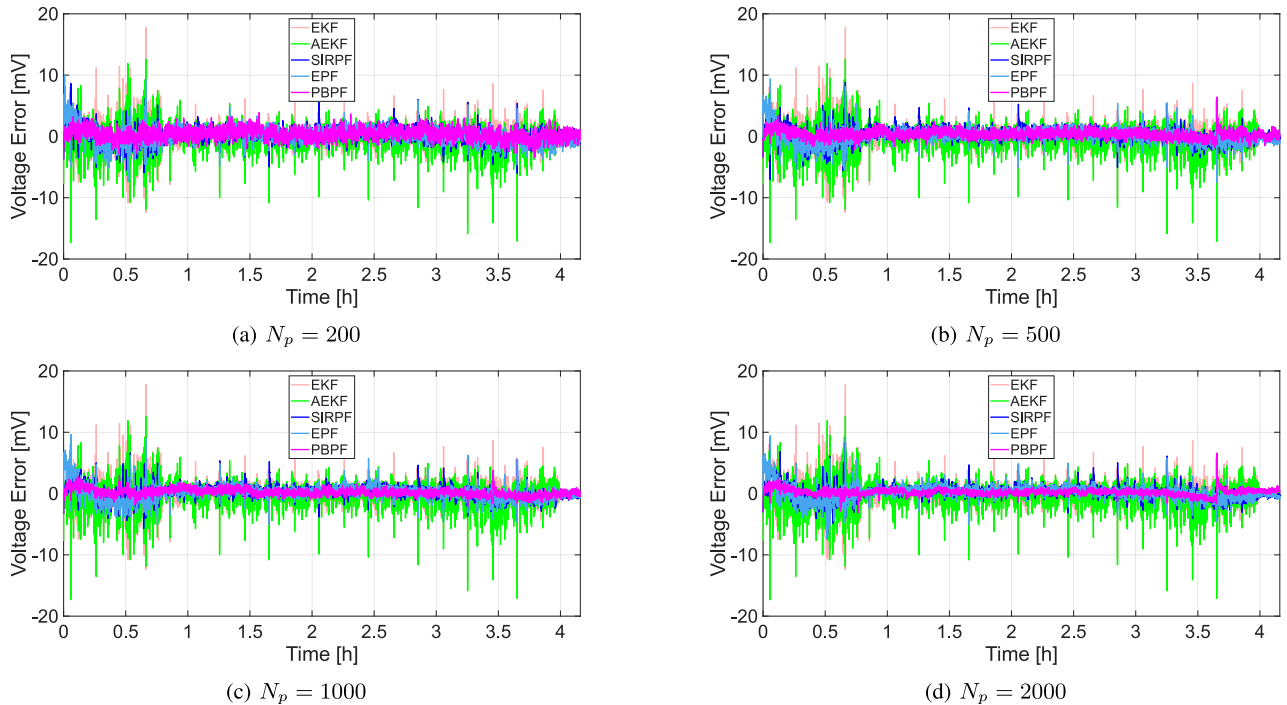


FIGURE 5. Terminal voltage estimation error with varying particle numbers for the LA92 driving cycle at 40 °C.

distribution; however, noticeable estimation deviations from the true SOC remain under dynamic operating conditions. As the particle count increases in cases (b)–(d), the estimation accuracy of both EPF and PBPF improves with increasing particle number, although the PBPF consistently provides the closest agreement with the true SOC. In contrast, the EKF

and AEKF exhibit stable behavior across all cases, serving as baseline references.

The corresponding SOC estimation errors are shown in Fig. 3, where the error is computed as the difference between the true SOC and the estimated SOC at each time step. In case (a), PBPF achieves a low average error with a relatively

higher variance due to the limited N_p . Compared with the conventional SIRPF, the EPF generally exhibits smoother error trajectories and reduced error fluctuations over several operating regions, although noticeable estimation errors remain under highly dynamic conditions. As the particle count increases in cases (b)–(d), the error variance of PBPF is progressively reduced, resulting in smoother and more stable estimation performance. The estimation accuracy of the EPF also benefits from increasing particle numbers; however, its error magnitude remains generally higher than that of the proposed PBPF. Although case (d) exhibits a slightly larger average error compared to case (a), the PBPF still maintains lower error magnitude and variance than the EPF, SIRPF and EKF-based methods, indicating superior robustness under dynamic operating conditions. Overall, the PBPF demonstrates a favorable trade-off between accuracy and robustness across different particle counts, outperforming the conventional SIRPF, EPF, and EKF-based methods, particularly at low and moderate N_p values.

2) TERMINAL VOLTAGE TRACKING PERFORMANCE

Accurate tracking of V_{term} plays a key role in SOC estimation, as it provides the primary measurement for state correction in model-based filtering frameworks. Errors in V_{term} estimation can directly propagate to SOC estimation errors, particularly under highly dynamic current conditions such as the LA92 driving cycle. In this subsection, the V_{term} (in V) is analyzed over time (in h) to assess the voltage tracking capability of each method.

Fig. 4 presents the V_{term} estimation results for all methods across different N_p . The proposed PBPF, shown in magenta, closely follows the measured voltage (black line) with minimal fluctuations. In case (a), a zoomed-in view between 2.8 and 3.2 h highlights that the PBPF remains well confined within the measured voltage envelope. The SIRPF exhibits noticeable oscillations, while the EKF and AEKF show larger deviations. The EPF provides smoother voltage tracking than the conventional SIRPF and follows the measured voltage more closely. However, the PBPF exhibits the closest agreement with the measured voltage, particularly during rapid voltage fluctuations. As the N_p value increases in cases (b)–(d), the voltage tracking performance of PBPF further improves, maintaining a tight fit to the measured voltage. The EPF also benefits from increasing particle numbers and exhibits lower variability than the conventional SIRPF. Nevertheless, the PBPF consistently achieves smaller voltage deviations than the EPF. Although the SIRPF benefits from increased N_p values, it continues to exhibit higher variability compared to the PBPF, while the EKFs remain more sensitive to dynamic voltage transients. These results indicate the superior robustness of PBPF in V_{term} estimation under dynamic operating conditions.

An analysis of the V_{term} errors is presented in Fig. 5, where the error is computed as the difference between the measured and estimated V_{term} . In case (a), the PBPF exhibits the lowest error magnitude, whereas the EKFs show noticeably

larger deviations. The EPF also reduces the voltage error compared with the conventional SIRPF. Its error fluctuations are generally smaller than those of the SIRPF but remain slightly larger than those of the PBPF. With increasing N_p values in cases (b)–(d), the error fluctuations of the PBPF become smoother and more consistent. The EPF also benefits from increasing particle numbers and produces more stable voltage error profiles than the conventional SIRPF. However, the PBPF consistently achieves lower RMSE, MAE, and MaxE values, indicating more accurate voltage estimation. Although SIRPF shows improved performance at higher N_p , its error fluctuations remain larger than those of the PBPF. These results confirm that the PBPF not only enhances SOC estimation accuracy but also provides robust and reliable V_{term} tracking under dynamic load conditions.

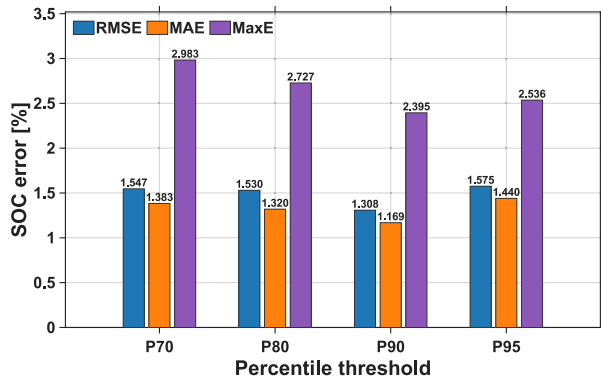
3) PERCENTILE SENSITIVITY ANALYSIS

To justify the selection of the cumulative-weight percentile threshold used in the proposed PBPF, a sensitivity analysis was conducted using the 70th, 80th, 90th, and 95th percentiles with a fixed particle number of $N_p = 200$ under the LA92 driving cycle at 40 °C. The corresponding SOC and V_{term} estimation results are presented in Fig. 6.

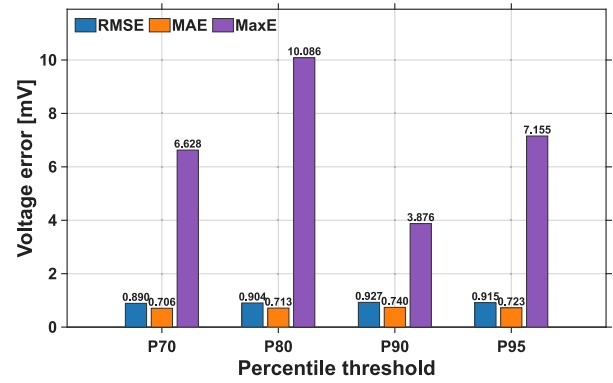
As shown in Fig. 6a, the percentile threshold has a noticeable influence on SOC estimation accuracy. Among the evaluated thresholds, the 90th percentile achieves the lowest RMSE (1.308%), MAE (1.169%), and MaxE (2.395%), providing the most balanced estimation performance. Lower percentile thresholds (70% and 80%) retain fewer high-weight particles, resulting in slightly larger estimation errors. In contrast, an excessively high threshold (95%) retains additional low-weight particles whose contributions are relatively small, slightly degrading the estimation accuracy.

A similar trend is observed for V_{term} estimation in Fig. 6b. The 90th percentile produces the lowest maximum voltage error (3.876 mV) while maintaining voltage RMSE (0.927 mV) and MAE (0.740 mV) comparable to those obtained using the other thresholds. Although the voltage RMSE and MAE vary only slightly across the four cases, the significantly reduced maximum voltage error demonstrates that the 90th percentile provides a more robust particle selection strategy.

Regarding computational efficiency, varying the percentile threshold has only a negligible effect on runtime because the overall algorithmic structure and computational complexity of the PBPF remain unchanged. The percentile threshold only changes the number of particles selected during the weighted resampling stage, while all remaining filtering operations are identical. Considering both estimation accuracy and computational efficiency, the 90th percentile provides the best overall trade-off for the LA92 driving cycle at 40 °C and is therefore adopted for this operating condition. For the additional validation cases at 25 °C and 0 °C, the optimal percentile is reselected based on the same sensitivity analysis procedure.



(a) SOC estimation accuracy



(b) Terminal voltage estimation accuracy

FIGURE 6. Sensitivity analysis of the proposed PBPF using different percentile thresholds (70th, 80th, 90th, and 95th percentiles) for $N_p = 200$ under the LA92 driving cycle at 40 °C. The corresponding SOC and terminal voltage estimation accuracy are shown in (a) and (b), respectively.

4) MONTE CARLO ANALYSIS

Since PFs are stochastic algorithms, a Monte Carlo (MC) analysis was conducted to evaluate the statistical robustness and repeatability of the proposed method. Ten independent runs were performed under the LA92 driving cycle at 40 °C for each particle number, and the corresponding estimation results are summarized in Table 3 as mean $\pm$ standard deviation.

As shown in Table 3, the proposed PBPF consistently achieves the lowest mean SOC RMSE, MAE, and MaxE compared with SIRPF and EPF for both $N_p = 200$ and $N_p = 500$. For example, at $N_p = 200$, PBPF reduces the mean SOC RMSE from 1.798% (SIRPF) and 1.770% (EPF) to 1.565%, while also achieving the lowest mean SOC MAE and MaxE. Similarly, at $N_p = 500$ it achieves a mean SOC RMSE of 1.616%, while also providing the lowest mean SOC MAE and MaxE, compared with 1.689% and 1.760% for SIRPF and EPF, respectively. A similar trend is observed for V_{term} estimation, where PBPF provides the lowest mean voltage RMSE and MAE while substantially reducing the maximum voltage error. Furthermore, the relatively small standard deviations obtained across the 10 independent runs indicate that the proposed PBPF exhibits stable and repeatable estimation performance despite the inherent randomness of particle filtering. These results demonstrate that the superiority of the proposed PBPF is statistically consistent rather than being attributable to a particular random realization.

B. RESULTS ON NASA B0005 CELL: 10TH DISCHARGE CYCLE

This subsection presents the SOC and V_{term} estimation results obtained using the NASA B0005 cell during its 10th discharge cycle to validate the proposed method. A dual filtering framework is employed, where the first filter (EKF, AEKF, SIRPF, EPF, and the proposed PBPF) estimates SOC and V_{term} , while a second EKF performs online estimation of unknown model parameters, including internal resistance and battery capacity, under dynamic discharge conditions.

The 10th discharge cycle is selected as it represents practical early-life uncertainty, where battery parameters can deviate from nominal values. The NASA dataset is adopted due to its wide use as a benchmark and the absence of explicitly provided model parameters, making it suitable for assessing estimation robustness. In this work, the discussion and analysis focus primarily on SOC and V_{term} estimation performance using the first filters, while details of the model parameter estimation process can be found in [40]; therefore, the parameter estimation results are not explicitly reported. Estimation accuracy is evaluated using SOC and V_{term} related error metrics. The dataset consists of 789 samples, which are converted into minutes for analysis.

All experiments use identical measurement inputs, and consistent noise covariance settings are maintained within each filter family so that performance differences reflect the filtering strategies rather than tuning advantages. The estimation performance is visualized through two figures showing SOC estimation (Fig. 7) and V_{term} estimation (Fig. 8). In both figures, the true SOC or the measured V_{term} is represented by a black solid line, whereas the DEKF, AEKF-EKF, SIRPF-EKF, EPF-EKF, and proposed PBPF-EKF results are indicated using light red dots, green squares, dark blue circles, light blue triangles, and magenta diamonds, respectively.

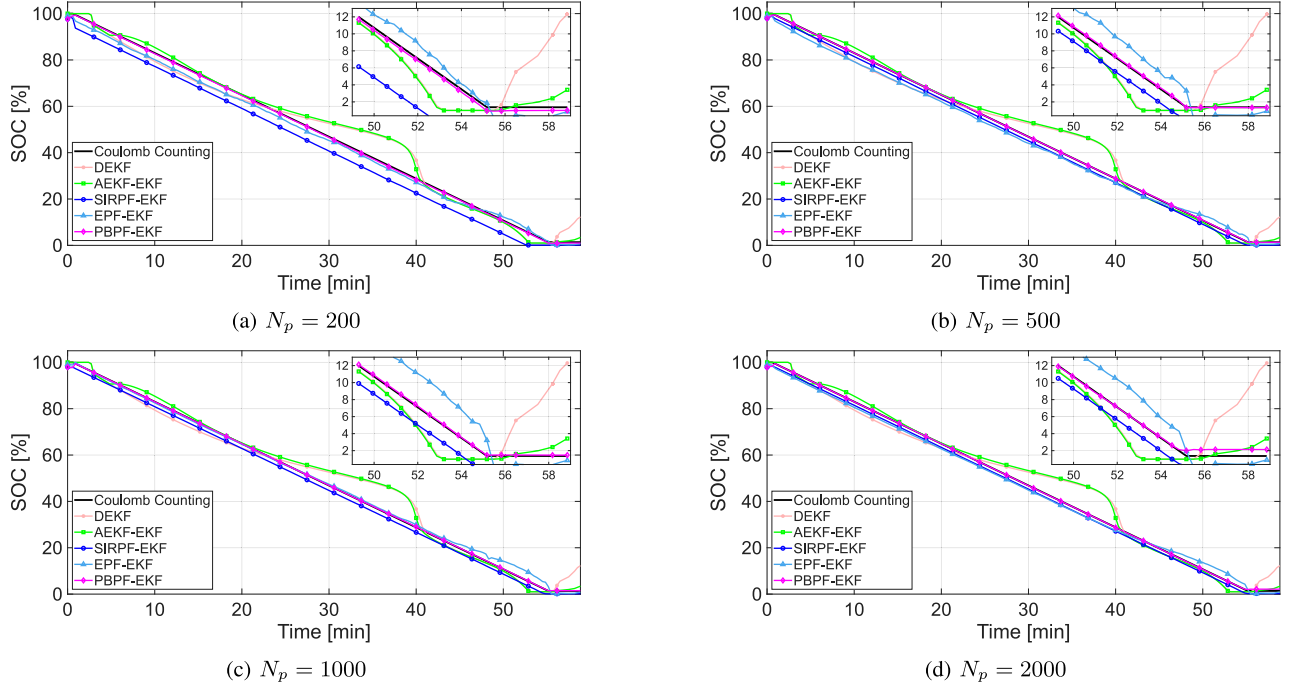
For PF-based approaches, including SIRPF-EKF, EPF-EKF, and PBPF-EKF, each figure further illustrates the effect of different particle numbers on SOC and V_{term} estimation, with (a) $N_p = 200$, (b) $N_p = 500$, (c) $N_p = 1000$, and (d) $N_p = 2000$.

1) SOC TRACKING PERFORMANCE

Accurate SOC tracking for the NASA B0005 cell during the 10th discharge cycle is particularly important, as battery model parameters are not explicitly provided and can already deviate from their nominal values, making this scenario representative of practical early-life uncertainty encountered in real-world BMS.

TABLE 3. Monte Carlo evaluation (10 independent runs) of SOC and terminal voltage estimation performance under the LA92 driving cycle at 40 °C. Results are reported as mean $\pm$ standard deviation.

N_p	Method	SOC RMSE (%)	SOC MAE (%)	SOC MaxE (%)	V_{term} RMSE (mV)	V_{term} MAE (mV)	V_{term} MaxE (mV)
200	SIRPF	1.798 ± 0.281	1.546 ± 0.263	3.723 ± 0.711	1.144 ± 0.051	0.831 ± 0.047	11.514 ± 0.921
	EPF	1.770 ± 0.198	1.553 ± 0.178	3.394 ± 0.423	1.238 ± 0.028	0.906 ± 0.027	10.698 ± 2.098
	PBPF	1.565 ± 0.190	1.379 ± 0.167	3.033 ± 0.343	0.952 ± 0.023	0.752 ± 0.019	5.984 ± 1.844
500	SIRPF	1.689 ± 0.285	1.450 ± 0.255	3.052 ± 0.657	1.042 ± 0.026	0.742 ± 0.024	11.011 ± 1.581
	EPF	1.760 ± 0.118	1.585 ± 0.109	3.140 ± 0.102	1.158 ± 0.025	0.834 ± 0.026	9.141 ± 0.642
	PBPF	1.616 ± 0.172	1.425 ± 0.130	2.904 ± 0.373	0.740 ± 0.050	0.583 ± 0.042	4.954 ± 1.328

**FIGURE 7.** SOC estimation for the NASA B0005 cell (10th discharge cycle) under varying particle numbers.

In Fig. 7, case (a) shows that the proposed PBPF-EKF, represented by the magenta line, achieves highly accurate SOC estimation. A zoomed-in view over the interval 49–59 min reveals that the PBPF-EKF closely follows the true SOC, whereas the conventional methods exhibit noticeable deviations. The EPF-EKF also improves the SOC tracking performance compared with the conventional SIRPF-EKF in the low-particle case ($N_p = 200$). However, it still exhibits larger deviations from the true SOC than the PBPF-EKF, particularly near the end of the discharge process. In case (b), the estimation performance of the SIRPF-EKF improves with an increased number of particles, and both the SIRPF-EKF and the PBPF-EKF closely follow the true SOC. In cases (b) and (c), the SIRPF-EKF and PBPF-EKF exhibit similar SOC trajectories, whereas the EPF-EKF shows slightly larger deviations from the true SOC. In case (d), the PBPF-EKF slightly overestimates the SOC within the interval 55–59 min, while it slightly underestimates the SOC in case (a) over the same interval; nevertheless, it provides the best overall SOC tracking performance among the compared approaches. Overall, these results highlight the superior robustness and accuracy of

the proposed PBPF-EKF method over the DEKF, AEKF-EKF, SIRPF-EKF, and EPF-EKF under parameter-uncertain conditions.

2) TERMINAL VOLTAGE TRACKING PERFORMANCE

Similar to the LA92 dataset, accurate V_{term} tracking is also essential for reliable SOC estimation in the NASA B0005 cell. Fig. 8 presents the V_{term} estimation results. In case (a), the proposed PBPF-EKF method achieves the most accurate V_{term} estimation among the EKF-based methods. The EPF-EKF also closely follows the measured voltage and exhibits performance comparable to that of the SIRPF-EKF. A zoomed-in view over the interval 30–50 min clearly indicates that the PF-based methods outperform the EKF-based methods in V_{term} tracking.

Although the PF-based approaches show nearly identical V_{term} estimation accuracy, the proposed PBPF-EKF continues to deliver superior SOC estimation performance compared with the SIRPF-EKF and EPF-EKF. This improvement can be attributed to the top-percentile weight selection strategy employed in the PBPF-EKF, which preserves high-quality particle weights and enhances SOC estimation

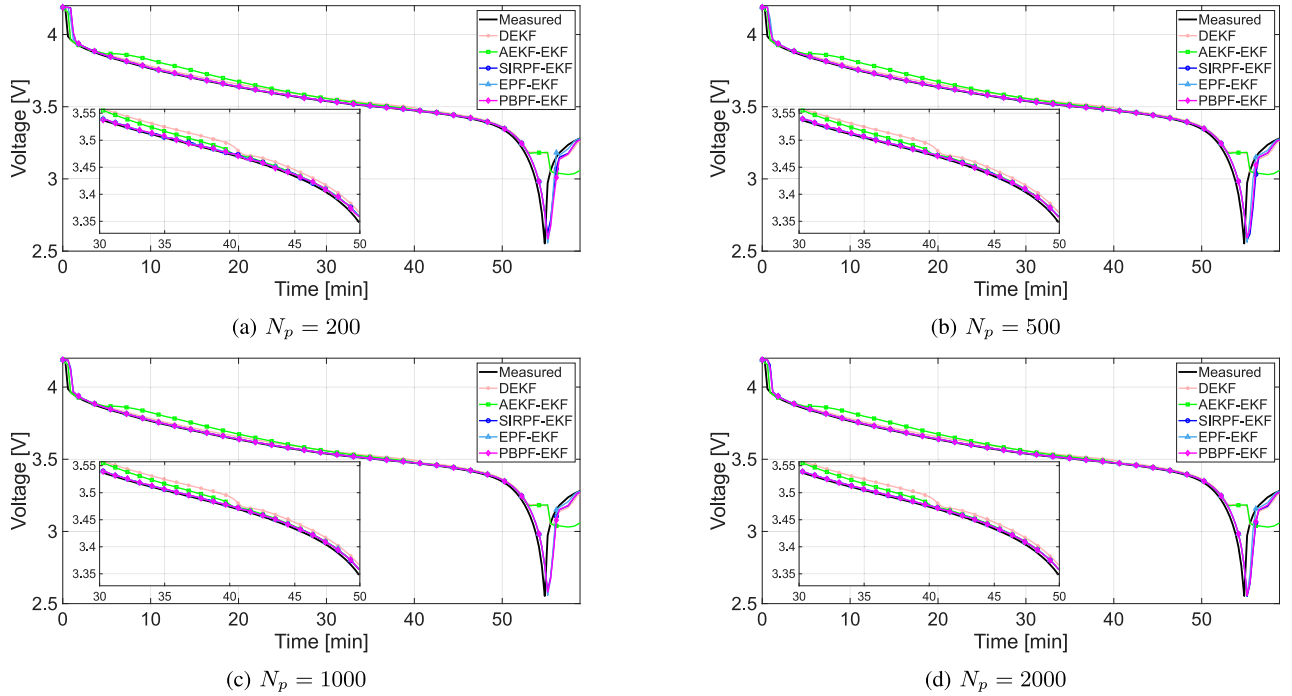


FIGURE 8. Terminal voltage estimation for the NASA B0005 cell (10th discharge cycle) under varying particle numbers.

accuracy beyond the standard SIRPF framework. In case (b), the PF-based methods again outperform the EKF-based methods, and similar trends are observed in cases (c) and (d). These results demonstrate that PF-based frameworks provide more consistent and accurate V_{term} tracking than EKF-based frameworks.

While the PBPF-EKF, SIRPF-EKF, and EPF-EKF exhibit comparable V_{term} tracking performance, the proposed PBPF-EKF achieves the best overall SOC estimation performance while maintaining competitive V_{term} estimation accuracy. This demonstrates the effectiveness of the proposed top-percentile particle weight selection strategy in improving the overall estimation performance.

C. GENERALIZATION UNDER ADDITIONAL OPERATING CONDITIONS

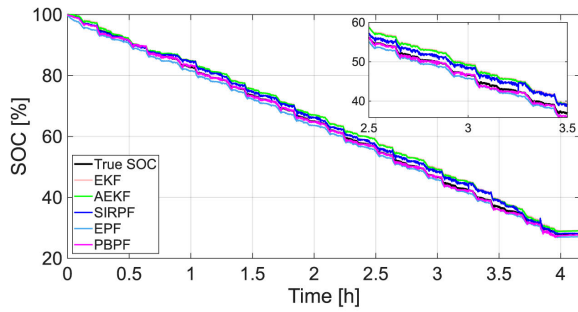
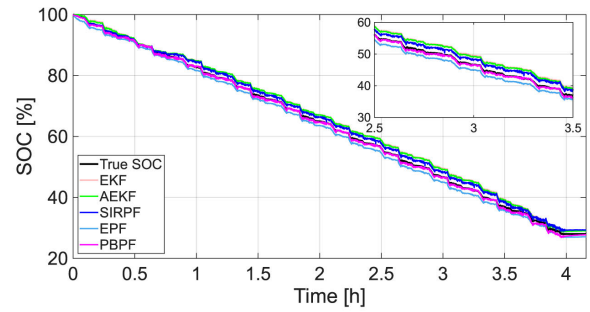
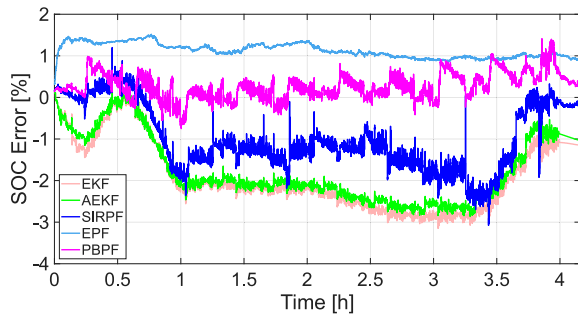
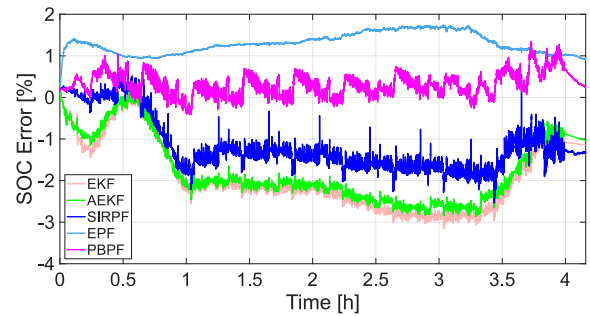
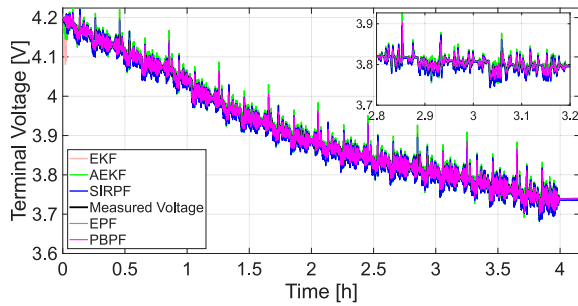
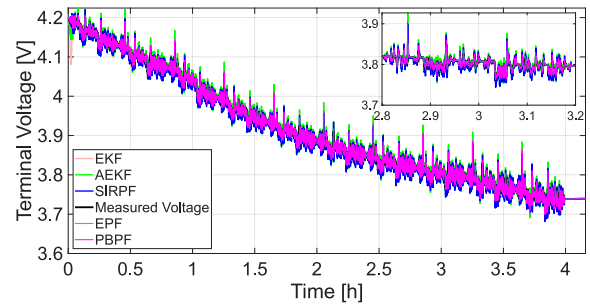
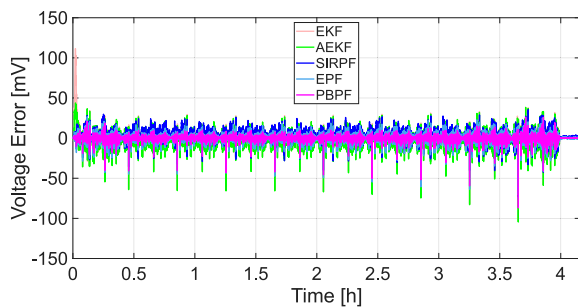
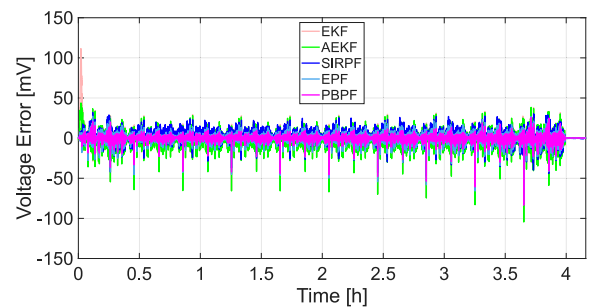
To further evaluate the robustness and generalization capability of the proposed PBPF, additional experiments are conducted under different operating temperatures. Specifically, the LA92 driving cycle is evaluated at 25 °C and 0 °C [38]. These additional experiments assess the performance of the proposed method beyond the primary benchmark case and demonstrate its effectiveness over a wider operating temperature range. For clarity, only the representative cases of $N_p = 200$ and $N_p = 2000$ are graphically presented, while the corresponding quantitative results for $N_p = 500$ and $N_p = 1000$ are included in the quantitative comparison presented in the following subsection.

1) LA92 DRIVING CYCLE AT 25 °C

Figs. 9 and 10 present the SOC tracking and corresponding estimation errors for the LA92 driving cycle at 25 °C. The

SOC error is defined as the difference between the true SOC and the estimated SOC; therefore, positive and negative errors indicate SOC underestimation and overestimation, respectively. At $N_p = 200$, the proposed PBPF closely follows the true SOC throughout the LA92 driving cycle and maintains its SOC error predominantly around zero despite the frequent load variations. In comparison, EKF and AEKF generally overestimate the true SOC, while SIRPF exhibits a smaller but still noticeable overestimation. The EPF produces a relatively smooth estimation trajectory but consistently underestimates the true SOC. A similar trend is observed for $N_p = 2000$, where PBPF continues to provide the closest agreement with the true SOC. Its error remains concentrated near zero throughout most of the driving cycle, whereas the systematic positive error of EPF and the negative errors of SIRPF, EKF, and AEKF remain evident.

The V_{term} estimation and corresponding voltage errors are presented in Figs. 11 and 12, respectively. All methods successfully reproduce the overall V_{term} profile and capture the rapid voltage variations induced by the dynamic current profile. Nevertheless, the enlarged views show that the proposed PBPF remains more closely aligned with the measured V_{term} and generally exhibits smaller deviations than the baseline methods. At $N_p = 200$, the PBPF voltage error remains concentrated around zero with a relatively narrow error envelope, whereas EKF, AEKF, and SIRPF exhibit more pronounced positive and negative voltage spikes. EPF also provides accurate V_{term} tracking with a relatively small error envelope; however, its SOC estimation accuracy remains inferior to that of PBPF. A similar trend is observed for $N_p = 2000$, where PBPF continues to maintain close agreement with the measured V_{term} , while its voltage error remains

(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 9.** SOC estimation with varying particle numbers for the LA92 driving cycle at 25 °C.(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 10.** SOC estimation error with varying particle numbers for the LA92 driving cycle at 25 °C.(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 11.** Terminal voltage estimation with varying particle numbers for the LA92 driving cycle at 25 °C.(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 12.** Terminal voltage estimation error with varying particle numbers for the LA92 driving cycle at 25 °C.

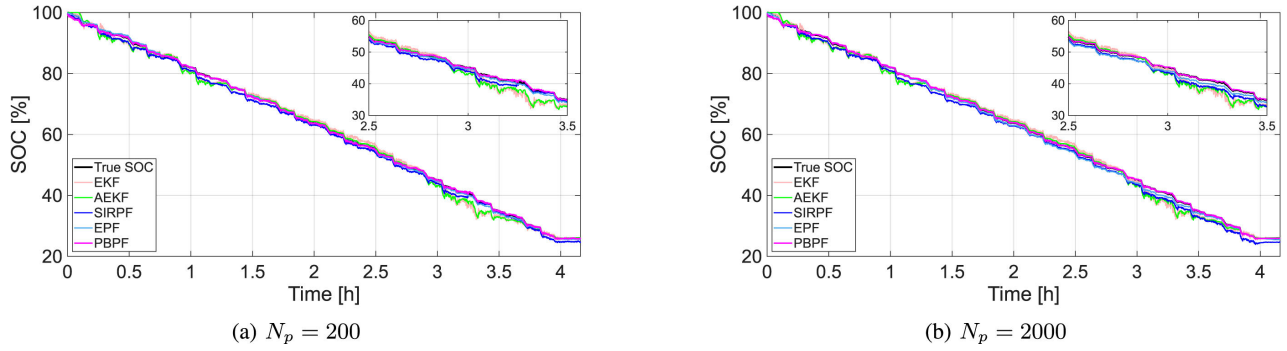


FIGURE 13. SOC estimation with varying particle numbers for the LA92 driving cycle at 0°C.

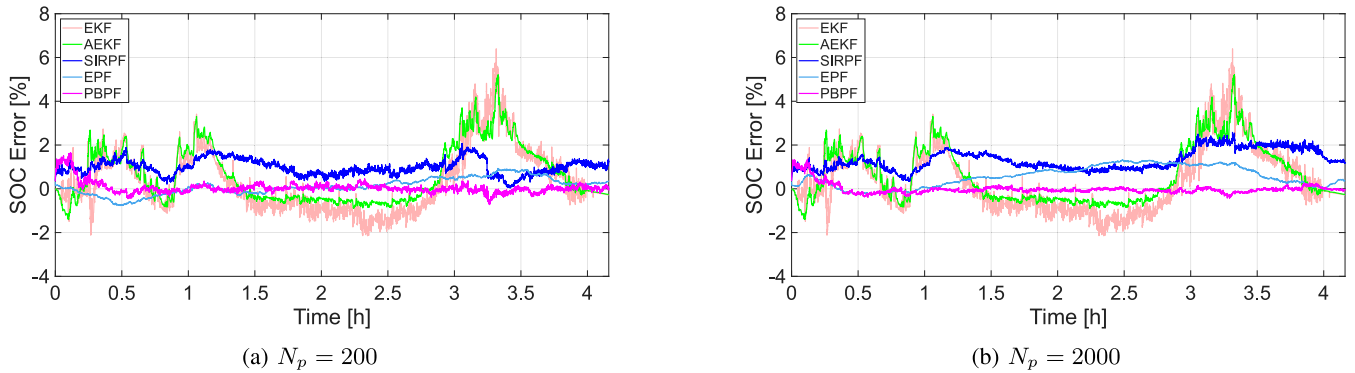


FIGURE 14. SOC estimation error with varying particle numbers for the LA92 driving cycle at 0°C.

centered around zero throughout most of the LA92 driving cycle.

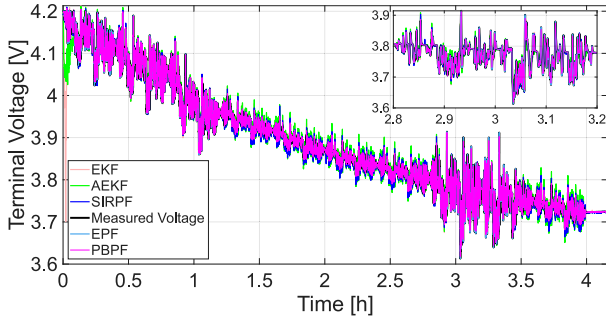
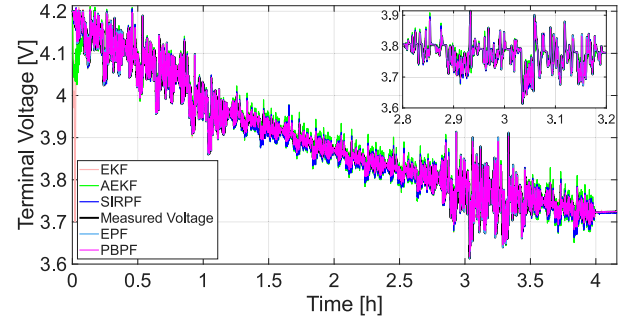
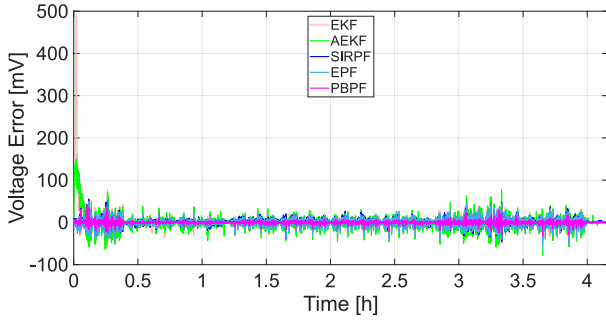
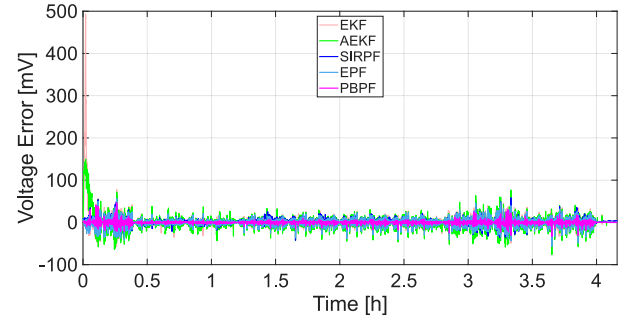
Overall, the graphical results consistently demonstrate that the proposed PBPF achieves the most balanced overall estimation performance by simultaneously providing accurate SOC estimation and V_{term} reconstruction under the LA92 driving cycle at 25°C. Across both representative particle numbers, PBPF closely follows the true SOC with an error predominantly concentrated around zero while simultaneously maintaining close agreement with the measured V_{term} and a relatively narrow voltage error envelope. Although EPF provides accurate V_{term} tracking, its systematic SOC underestimation remains evident throughout the driving cycle. In contrast, EKF, AEKF, and SIRPF exhibit larger systematic SOC deviations together with more pronounced voltage error spikes, indicating lower estimation robustness under the dynamic operating conditions of the LA92 driving cycle. These graphical observations are further supported by the quantitative comparison presented in the following subsection, which reports the complete RMSE, MAE, and MaxE results for $N_p \in \{200, 500, 1000, 2000\}$.

2) LA92 DRIVING CYCLE AT 0°C

Figs. 13 and 14 illustrate the SOC estimation results and corresponding error profiles for the LA92 driving cycle at 0°C. Under the representative case of $N_p = 200$, the proposed PBPF accurately tracks the true SOC over the entire driving cycle, with its estimation error remaining tightly

distributed around zero. In contrast, EKF and AEKF exhibit noticeably larger error fluctuations, whereas SIRPF shows a persistent tendency to underestimate the true SOC. EPF also achieves satisfactory SOC tracking with relatively small estimation errors. At $N_p = 2000$, PBPF continues to maintain close agreement with the true SOC and preserves a stable error profile throughout most of the driving cycle. Although EPF remains a competitive baseline, its SOC estimation accuracy deteriorates at $N_p = 2000$ compared with the $N_p = 200$ case, whereas PBPF exhibits more consistent estimation performance across both representative particle numbers. The systematic underestimation of SIRPF together with the larger error fluctuations of EKF and AEKF also remain evident.

Figs. 15 and 16 depict the estimated V_{term} and the associated voltage errors for the LA92 driving cycle at 0°C. All methods successfully reconstruct the overall V_{term} profile, while the enlarged views highlight noticeable differences in estimation accuracy. Throughout both representative particle numbers, PBPF maintains the closest agreement with the measured V_{term} . At $N_p = 200$, its voltage error remains tightly centered around zero, whereas EKF, AEKF, and SIRPF exhibit larger fluctuations, especially during rapid load transients. EPF also achieves accurate voltage tracking but produces slightly larger deviations than PBPF. The same trend is observed for $N_p = 2000$, where PBPF continues to provide stable and consistent voltage reconstruction with only small deviations from the measured V_{term} . Although EPF also

(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 15.** Terminal voltage estimation with varying particle numbers for the LA92 driving cycle at 0 °C.(a) $N_p = 200$ (b) $N_p = 2000$ **FIGURE 16.** Terminal voltage estimation error with varying particle numbers for the LA92 driving cycle at 0 °C.

achieves accurate voltage tracking, its voltage errors remain consistently larger than those of PBPF.

The graphical comparisons suggest that the proposed PBPF preserves stable estimation performance under the LA92 driving cycle at 0 °C, regardless of the representative particle number considered. The SOC and V_{term} estimates remain in close agreement with the corresponding reference measurements, while the associated estimation errors stay confined to a relatively narrow range. Although EPF also provides satisfactory estimation accuracy, its performance varies more noticeably between $N_p = 200$ and $N_p = 2000$, whereas PBPF maintains consistent estimation quality across both cases. Conversely, EKF, AEKF, and SIRPF exhibit larger estimation deviations and error fluctuations throughout the driving cycle. These graphical findings are further substantiated by the quantitative comparison in the following subsection, where the complete RMSE, MAE, and MaxE results for $N_p \in \{200, 500, 1000, 2000\}$ are presented.

D. COMPARATIVE ANALYSIS

To ensure a fair comparison, all error metrics are evaluated under identical experimental settings for each dataset. Quantitative performance is assessed using the RMSE, MAE, and MaxE for both SOC and V_{term} , considering the LA92 driving cycle at 40 °C and the NASA B0005 cell at the 10th discharge cycle. The results are summarized in Fig. 17. Cases (a) and (d) present the RMSE and MAE of SOC estimation

for the LA92 and NASA datasets, respectively. In case (a), the proposed PBPF (magenta bars) consistently exhibits lower SOC estimation errors than the conventional methods, including EKF (red), AEKF (green), SIRPF (dark blue), and EPF (light blue). The EPF generally provides comparable SOC estimation accuracy to the conventional SIRPF but remains less accurate than the proposed PBPF. Although increasing N_p slightly increases the error in PF-based methods, the PBPF still outperforms the other approaches. This behavior reflects the inherent difficulty of tuning noise covariance parameters in PF-based estimators. For the NASA dataset, as shown in case (d), the proposed PBPF-EKF achieves both RMSE and MAE SOC errors below 0.34% across all N_p values. The EPF-EKF also substantially reduces the SOC estimation errors compared with the DEKF and AEKF-EKF, although its errors remain higher than those of the PBPF-EKF. In contrast, the SIRPF-EKF shows higher errors at $N_p = 200$ and SOC RMSE and MAE values ranging from approximately 1.5% to 2.1% for the remaining particle counts, while the DEKF and AEKF-EKF exhibit SOC RMSE and MAE values of approximately 5% or below.

The corresponding RMSE and MAE results for V_{term} are shown in cases (b) and (e) for the LA92 and NASA datasets, respectively. As illustrated in case (b), the proposed PBPF exhibits decreasing V_{term} estimation errors with increasing N_p , achieving the best performance with RMSE and MAE values below 1 mV. The EPF provides comparable V_{term} estimation accuracy to the conventional SIRPF, while

the proposed PBPF achieves the highest V_{term} estimation accuracy for the LA92 dataset. In case (e), the SIRPF-EKF, EPF-EKF, and PBPF-EKF exhibit comparable V_{term} estimation accuracy and significantly outperform the EKF-based methods. Specifically, PBPF-EKF achieves an RMSE of approximately 55 mV, whereas both the SIRPF-EKF and EPF-EKF achieve RMSE values below 60 mV. The SIRPF-EKF, EPF-EKF, and PBPF-EKF all maintain MAE values of approximately 16.1 mV or below. Among the PF-based methods, the EPF-EKF achieves the lowest V_{term} RMSE and MAE for the NASA dataset, whereas the PBPF-EKF provides the best overall SOC estimation performance.

Furthermore, case (c) reports the MaxE values for SOC and V_{term} under the LA92 cycle. The proposed PBPF clearly achieves the lowest maximum errors, with SOC MaxE below 2.64% and V_{term} MaxE below 6.61 mV, significantly outperforming the traditional filtering methods. Compared with the conventional SIRPF, the EPF generally exhibits higher maximum SOC and V_{term} errors, whereas the PBPF consistently achieves the lowest maximum errors for both SOC and V_{term} .

Finally, case (f) summarizes the overall improvement relative to the EKF baseline for both SOC and V_{term} estimation. The improvement is computed as the average of RMSE, MAE, and MaxE for the LA92 dataset and the average of RMSE and MAE for the NASA cell. For each metric, the values are first averaged over all tested particle numbers $N_p \in \{200, 500, 1000, 2000\}$, and all metrics are equally weighted. In the plot, orange bars correspond to the LA92 (40°C) results, while blue bars represent the NASA B0005 (10th discharge cycle) case. The proposed PBPF achieves the highest overall improvement in SOC estimation for both datasets and also provides the largest V_{term} improvement for the LA92 dataset. For the NASA dataset, the EPF achieves the largest improvement in V_{term} estimation, while the PBPF remains the best overall method owing to its superior SOC estimation performance. The AEKF shows only marginal gains over the EKF in SOC estimation and inferior performance in V_{term} estimation. These results confirm that the proposed PBPF delivers the most consistent and significant overall performance enhancement over the EKF across both driving-cycle test scenarios. The detailed RMSE, MAE, MaxE, and overall improvement results for the LA92 dataset at 40°C and the NASA B0005 battery at the 10th discharge cycle are presented in Tables 4 and 5, respectively, and are further discussed in the following subsection.

E. QUANTITATIVE COMPARISON

Table 4 presents the quantitative comparison of the investigated methods for the LA92 driving cycle at 40°C, 25°C, and 0°C over different particle numbers. While the quantitative results at 40°C have already been discussed in the previous subsection, this table further examines the estimation performance under additional operating temperatures. A similar trend is observed across all temperature conditions, where

the proposed PBPF consistently exhibits the lowest overall SOC and V_{term} estimation errors together with the highest overall improvement relative to the EKF baseline. For example, at 0°C and $N_p = 2000$, the proposed PBPF achieves the lowest SOC RMSE of 0.236% and the highest overall SOC improvement of 83.57%, demonstrating its ability to maintain accurate SOC estimation even under low-temperature operation. Although EPF generally improves the SOC estimation accuracy over the conventional SIRPF, its SOC estimation performance varies more noticeably with particle number and operating temperature than that of the proposed PBPF. These quantitative results further confirm the effectiveness and robustness of the proposed PBPF in SOC estimation over a broad range of operating conditions.

Furthermore, Table 5 summarizes the quantitative results for the NASA B0005 battery at discharge Cycle 10 under different particle numbers. Consistent with the LA92 results, the proposed PBPF achieves the highest SOC estimation accuracy and the largest overall improvement relative to the DEKF baseline. For instance, at $N_p = 500$, the proposed PBPF attains the lowest SOC RMSE of 0.07% together with the highest overall SOC improvement of 98.66%. Overall, these quantitative results further demonstrate the effectiveness and robustness of the proposed PBPF for accurate SOC estimation across different datasets, operating conditions, and particle numbers.

F. COMPUTATIONAL COMPLEXITY ANALYSIS

The proposed PBPF employs the PBR scheme introduced in [28], which aims to reduce the computational burden associated with conventional SIR resampling while maintaining comparable or improved tracking performance. This is achieved by selecting a subset of high-weight parent particles, where the number of selected particles N_{perc} is smaller than the total number of particles N_p . New offspring particles are then generated proportionally to the parent weights while keeping N_p unchanged. As a result, the PBPF provides an efficient balance between computational cost and estimation accuracy, making it suitable for real-time SOC and V_{term} estimation in BMS applications.

The theoretical computational complexities of the compared estimators are summarized in Table 6. For a state dimension of n_x , the EKF and AEKF require covariance propagation and update operations with a computational complexity of $\mathcal{O}(n_x^3)$ per time step. The SIRPF and PBPF propagate and evaluate all N_p particles, resulting in a regular filtering complexity of $\mathcal{O}(N_p n_x)$. The EPF additionally performs an EKF correction for every particle and therefore requires $\mathcal{O}(N_p n_x^3)$ operations per time step. Standard systematic resampling can be implemented with a complexity of $\mathcal{O}(N_p)$, whereas the proposed PBR strategy requires particle sorting and consequently has a resampling complexity of $\mathcal{O}(N_p \log N_p)$. Since the state dimension is fixed at $n_x = 3$ in this study, the EKF and AEKF have constant complexity with

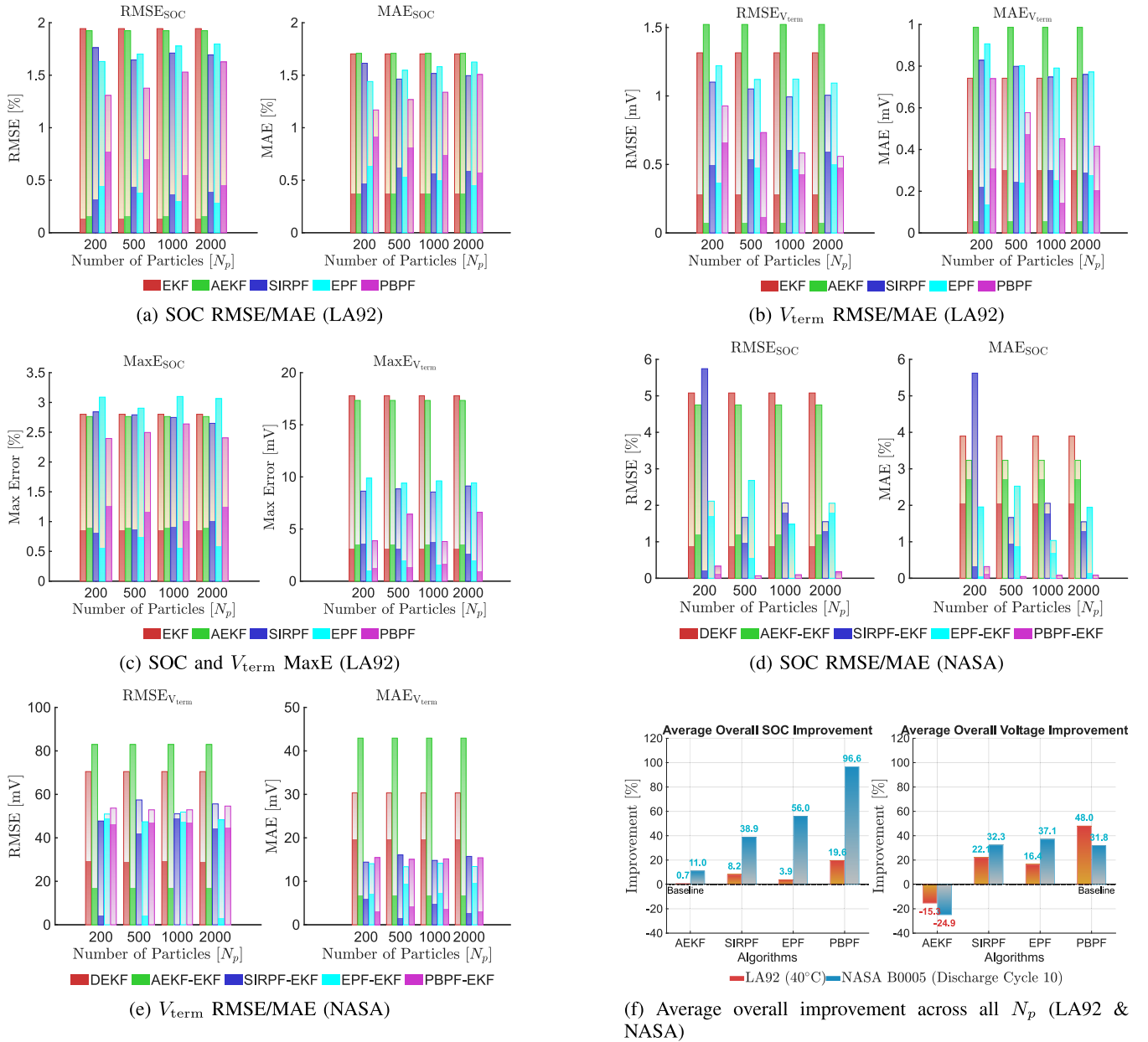


FIGURE 17. SOC and terminal voltage error metrics for the LA92 (40°C) and NASA B0005 (10th discharge cycle) datasets.

respect to N_p , while the dominant regular filtering costs of the PF-based methods scale linearly with the particle number.

The theoretical complexity analysis is further validated experimentally using the runtime results shown in Fig. 18. As expected, the EKF and AEKF exhibit nearly constant runtimes because their computational cost is independent of the particle number, requiring only 0.039–0.044 ms per step for the LA92 dataset and 0.1136–0.2132 ms per step for the NASA B0005 dataset. In contrast, the runtimes of the PF-based methods increase approximately linearly with N_p for both datasets. Among the PF-based methods, the EPF incurs the highest computational cost, with runtime per step increasing from 13.41 to 131.098 ms for LA92 and from 3.9901 to 26.0438 ms for NASA B0005 as N_p increases

from 200 to 2000. The proposed PBPF consistently requires substantially less runtime than the EPF while maintaining runtime comparable to that of the conventional SIRPF. For example, at $N_p = 2000$, the PBPF requires 95.994 ms per step compared with 131.098 ms for the EPF on the LA92 dataset, and 3.6171 ms compared with 26.0438 ms on the NASA B0005 dataset. Notably, for the NASA B0005 dataset, the PBPF even achieves lower runtime than the SIRPF at moderate and high particle numbers ($N_p \geq 1000$), requiring 1.7731 and 3.6171 ms per step compared with 2.1989 and 4.7086 ms for the SIRPF at $N_p = 1000$ and 2000, respectively. These results demonstrate that the proposed PBPF achieves an effective tradeoff between estimation accuracy and computational cost, making it well suited for

TABLE 4. Comparison of SOC and terminal voltage estimation performance for the LA92 driving cycle under different temperatures and particle numbers.

Temp. (°C)	Method	N_p	SOC Error (%)			V_{term} Error (mV)			Overall Improvement over EKF (%)	
			RMSE	MAE	MaxE	RMSE	MAE	MaxE	SOC	V_{term}
40	EKF	–	1.944	1.703	2.801	1.315	0.742	17.789	–	–
	AEKF	–	1.924	1.708	2.763	1.522	0.986	17.335	0.70	-15.32
	SIRPF	200	1.762	1.614	2.842	1.101	0.828	8.633	4.38	18.75
		500	1.645	1.463	2.792	1.051	0.799	8.852	9.95	20.91
		1000	1.711	1.517	2.748	0.993	0.748	8.549	8.27	25.25
		2000	1.693	1.496	2.649	1.005	0.760	9.129	10.15	23.31
	EPF	200	1.631	1.440	3.089	1.221	0.907	9.879	7.08	9.83
		500	1.701	1.547	2.904	1.120	0.802	9.398	6.00	18.00
		1000	1.778	1.581	3.100	1.123	0.791	9.622	1.67	18.01
		2000	1.797	1.626	3.063	1.094	0.772	9.423	0.91	19.95
	PBPF	200	1.308	1.169	2.395	0.927	0.740	3.876	26.19	36.02
		500	1.377	1.268	2.496	0.733	0.577	6.421	21.87	43.48
		1000	1.530	1.338	2.639	0.584	0.452	3.802	16.16	57.77
		2000	1.627	1.507	2.405	0.559	0.416	6.606	13.99	54.80
25	EKF	–	2.071	1.900	3.149	8.989	4.776	111.541	–	–
	AEKF	–	1.928	1.756	2.865	7.707	4.767	104.528	7.84	6.91
	SIRPF	200	1.285	1.089	3.077	6.810	4.873	73.132	27.65	18.88
		500	1.372	1.248	2.525	6.595	4.556	89.372	29.31	17.04
		1000	1.283	1.162	2.768	6.673	4.658	83.092	29.66	17.92
		2000	1.356	1.231	2.535	6.606	4.591	74.826	29.76	21.10
	EPF	200	1.139	1.127	1.518	4.411	2.658	70.856	45.83	43.91
		500	0.884	0.878	1.121	4.381	2.575	80.959	58.52	41.59
		1000	1.137	1.121	1.550	4.372	2.558	80.268	45.63	41.95
		2000	1.323	1.300	1.739	4.372	2.554	80.631	37.50	41.86
	PBPF	200	0.406	0.319	1.413	3.083	1.689	86.932	72.92	50.80
		500	0.434	0.355	1.133	3.064	1.685	83.918	74.79	51.80
		1000	0.412	0.317	1.396	3.053	1.675	81.953	73.04	52.50
		2000	0.405	0.325	1.343	3.061	1.682	83.601	73.57	51.93
0	EKF	–	1.472	1.142	6.397	21.592	6.103	493.263	–	–
	AEKF	–	1.415	1.056	5.210	16.056	7.719	149.731	9.96	22.93
	SIRPF	200	1.041	0.980	2.077	6.306	4.217	54.637	36.99	63.54
		500	1.005	0.795	2.133	6.174	4.109	58.453	42.90	64.07
		1000	1.310	1.260	2.204	6.151	4.100	56.496	22.07	64.29
		2000	1.396	1.318	2.580	6.278	4.206	58.538	16.46	63.38
	EPF	200	0.393	0.308	0.933	6.193	3.819	56.446	77.24	65.77
		500	0.810	0.715	1.338	6.156	3.747	55.922	53.82	66.25
		1000	1.406	1.134	2.290	6.158	3.720	56.686	23.11	66.34
		2000	0.737	0.612	1.331	6.123	3.679	57.373	58.53	66.57
	PBPF	200	0.277	0.156	1.658	2.564	1.412	50.886	80.54	84.89
		500	0.291	0.161	1.795	2.605	1.371	50.174	79.37	85.10
		1000	0.282	0.183	1.539	2.547	1.352	47.545	80.27	85.47
		2000	0.236	0.139	1.349	2.558	1.345	46.547	83.57	85.56

Note: Overall improvement is calculated as the arithmetic mean of the percentage reductions in RMSE, MAE, and MaxE relative to the EKF baseline. Bold values indicate the best performance in each metric.

real-time battery management applications. It should be noted that the reported runtime values are hardware-dependent and can vary depending on the computing platform. Nevertheless, the relative computational trends and comparisons among the evaluated methods remain unchanged.

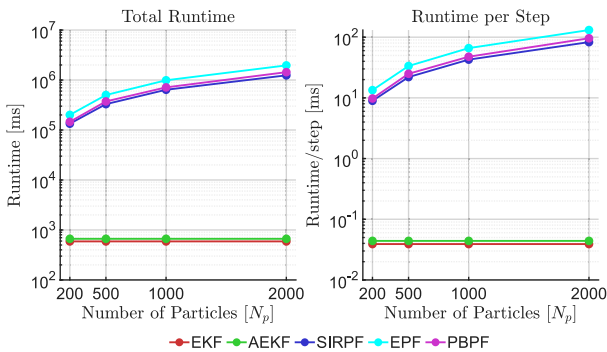
G. DISCUSSION ON ACCURACY-COMPLEXITY TRADE-OFF

Overall, the experimental results for the LA92 driving cycle at 40°C demonstrate that the proposed PBPF achieves the most favorable balance between estimation accuracy and computational efficiency under highly dynamic and

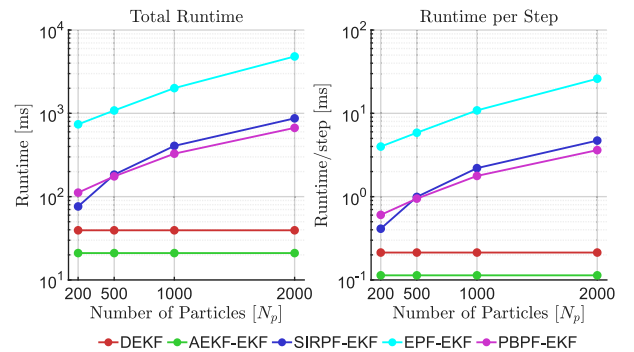
TABLE 5. Comparison of SOC and terminal voltage estimation performance for the NASA B0005 battery at discharge Cycle 10 under different particle numbers.

Cycle	Method	N_p	SOC Error (%)		V_{term} Error (mV)		Overall Improvement over DEKF (%)	
			RMSE	MAE	RMSE	MAE	SOC	V_{term}
10	DEKF	—	5.0785	3.8957	70.4785	30.3336	—	—
	AEKF	—	4.7501	3.2326	82.9489	42.9408	11.05	-24.88
	SIRPF	200	5.7442	5.6223	47.7	14.4	-26.66	38.40
		500	1.6695	1.6641	57.4	16.1	62.90	27.09
		1000	2.0645	2.0554	51.1	14.8	54.09	34.63
		2000	1.5513	1.5478	55.6	15.7	65.45	30.57
	EPF	200	2.11	1.95	51.0	14.1	54.76	35.43
		500	2.68	2.52	47.4	13.4	42.06	39.69
		1000	1.48	1.04	51.9	14.1	71.92	34.53
		2000	2.06	1.94	48.3	13.4	55.43	38.80
	PBPF	200	0.33	0.32	53.7	15.5	92.76	31.36
		500	0.07	0.05	52.9	15.0	98.66	32.65
		1000	0.10	0.08	52.9	15.2	97.99	32.45
		2000	0.18	0.08	54.6	15.4	97.10	30.57

Note: Overall improvement is calculated from the percentage reduction in the arithmetic mean of RMSE and MAE relative to the DEKF baseline, separately for SOC and terminal voltage estimation. Bold values indicate the best performance in each metric.



(a) Computational runtime comparison (LA92)



(b) Computational runtime comparison (NASA B0005)

FIGURE 18. Comparison of total runtime and runtime per step for different particle numbers on the LA92 and NASA B0005 datasets.**TABLE 6.** Theoretical computational complexity of the compared estimators per time step.

Method	Regular filtering operation	Resampling operation
EKF	$\mathcal{O}(n_x^3)$	—
AEKF	$\mathcal{O}(n_x^3)$	—
SIRPF	$\mathcal{O}(N_p n_x)$	$\mathcal{O}(N_p)$
EPF	$\mathcal{O}(N_p n_x^3)$	$\mathcal{O}(N_p)$
PBPF	$\mathcal{O}(N_p n_x)$	$\mathcal{O}(N_p \log N_p)$

parameter-uncertain operating conditions. The observed accuracy improvements are not solely due to enhanced V_{term} tracking, as the PBPF achieves lower SOC errors even when the V_{term} estimation accuracy is comparable to that of the SIRPF and EPF for the LA92 driving cycle at 40°C. Consequently, the EKF-based methods exhibit a gradual accumulation of SOC estimation bias under parameter uncertainty because they rely on local linearization and a

single Gaussian approximation of the posterior distribution. This effect becomes more pronounced during rapid changes in battery operating conditions, where modeling errors and linearization inaccuracies accumulate over time. In contrast, the PF-based methods maintain more consistent SOC tracking by preserving multiple weighted state hypotheses, providing greater robustness to nonlinearities and parameter uncertainties. The proposed PBPF further improves this robustness by mitigating particle impoverishment through the proposed PBR strategy. Since identical models, initialization settings, datasets, and noise covariances are used within each filter family, the observed performance differences primarily reflect the characteristics of the underlying filtering algorithms rather than differences in the battery model or tuning parameters.

The PBPF further demonstrates a favorable accuracy–complexity trade-off with respect to particle count. While

the SIRPF generally requires a larger number of particles to achieve satisfactory SOC estimation accuracy, the PBPF provides high estimation accuracy even at relatively low particle numbers (e.g., $N_p = 200$). For the LA92 driving cycle at 40 °C, increasing N_p beyond moderate values (around $N_p = 1000$) results in only marginal accuracy improvements relative to the additional computational cost, indicating an efficient operating region for practical implementations. At 25 °C, the proposed PBPF exhibits stable SOC and V_{term} estimation performance over the investigated particle numbers, with only minor variations in estimation accuracy. At 0 °C, the proposed PBPF likewise maintains robust estimation performance across all particle numbers, although the best overall SOC estimation accuracy is achieved at $N_p = 2000$. These results indicate that the proposed PBPF maintains a favorable accuracy–complexity trade-off over a wide range of operating temperatures.

It is worth noting that, as summarized in Table 7, the proposed PBPF consistently required fewer resampling events than both SIRPF and EPF under identical experimental settings. The reported resampling count denotes the total number of time steps during the entire drive cycle at which the resampling procedure was triggered, rather than the number of particles being resampled. The corresponding resampling rate is calculated as the percentage of the total filter iterations during the entire drive cycle in which the resampling procedure was triggered. Since resampling is triggered whenever the ESS falls below the predefined threshold (e.g., $0.3N_p$), the reduced resampling frequency suggests that the proposed PBPF weighting strategy is able to maintain an effective particle representation for a longer duration. Consequently, the number of unnecessary resampling events is reduced, which potentially helps mitigate the risk of particle impoverishment while preserving the superior SOC and V_{term} estimation performance of the proposed method.

TABLE 7. Comparison of resampling events under different particle numbers for the LA92 driving cycle at 40 °C. The resampling count represents the total number of time steps during the entire drive cycle at which the resampling procedure was triggered. The resampling rate is calculated as the percentage of the total 14,961 filter iterations during the drive cycle in which the resampling procedure was triggered.

Method	N_p	Resampling Count	Resampling Rate (%)
SIRPF	200	31	0.207
EPF	200	19	0.127
PBPF	200	11	0.074
SIRPF	500	16	0.107
EPF	500	16	0.107
PBPF	500	9	0.060
SIRPF	1000	14	0.094
EPF	1000	15	0.100
PBPF	1000	7	0.047
SIRPF	2000	11	0.074
EPF	2000	14	0.094
PBPF	2000	7	0.047

Finally, the dataset-specific results confirm the practical relevance of the PBPF: under the LA92 cycle, it maintains stable SOC and V_{term} tracking during rapid current transients

with reduced worst-case errors, while for the NASA B0005 cell, PBPF-EKF improves SOC estimation within a DEKF framework without modifying parameter estimation. By narrowing the performance gap between EKF- and PF-based estimators while requiring fewer particles, the PBPF represents a practical solution for real-time embedded BMS applications. Future work could explore adaptive percentile selection or dynamic particle allocation to further enhance performance under highly dynamic and uncertain conditions.

VI. CONCLUSION

This paper presented a PBPF algorithm for SOC and V_{term} estimation in LIBs, motivated by the limitations of conventional particle filtering methods, including particle degeneracy and high computational cost. Unlike standard PF approaches that refine the proposal density, PBPF directly targets the resampling stage by retaining particles according to percentile-ranked importance weights and allocating offspring proportionally. This strategy reduces weight degeneracy and particle impoverishment while capturing the essential characteristics of the posterior distribution using a substantially reduced particle set, maintaining high estimation accuracy for both SOC and V_{term} .

The proposed method was comprehensively validated using the LA92 driving cycle at 40 °C, 25 °C, and 0 °C, together with the NASA B0005 battery under the 10th discharge cycle. Comparative evaluations against EKF, AEKF, conventional SIRPF, and EPF demonstrate that the proposed PBPF consistently provides the best overall SOC estimation performance while maintaining competitive or superior V_{term} estimation accuracy across different particle numbers and operating conditions. Additional percentile sensitivity and Monte Carlo analyses further confirm the robustness, repeatability, and computational efficiency of the proposed percentile-based resampling strategy. Overall, these results demonstrate that PBPF provides a favorable accuracy–complexity trade-off, making it well suited for real-time embedded BMS applications.

From a broader perspective, PBPF has the potential to enhance the safety, efficiency, and longevity of EV battery packs. Future research could focus on:

- 1) **Adaptive percentile selection:** Dynamically adjusting the retained particle percentage according to system conditions, while investigating the effects of perc and particle diversity metrics (e.g., ESS or particle entropy) on SOC estimation accuracy and computational cost.
- 2) **Temperature- and aging-aware models:** Incorporating thermal effects and battery degradation models to improve SOC and V_{term} estimation under varying temperatures and aging conditions, with future extension to SOH estimation.
- 3) **Joint SOC and SOH estimation:** Extending PBPF to simultaneously estimate SOC and SOH for more comprehensive battery state monitoring.

- 4) **Hardware-in-the-loop validation:** Implementing PBPF on embedded BMS platforms to evaluate real-time performance across different battery chemistries and operating conditions.
- 5) **Comparison with advanced PF methods:** Extending the comparative study to representative PF variants and emerging particle filtering frameworks under broader operating scenarios.
- 6) **Expanded dataset testing:** Validating PBPF using additional driving cycles and aged battery datasets to further assess its scalability and long-term applicability.

Overall, the proposed PBPF demonstrates that improving the resampling mechanism, rather than solely refining the proposal density, can effectively enhance PF performance for battery SOC estimation. The proposed percentile-based resampling strategy provides a practical and computationally efficient solution for accurate SOC and V_{term} estimation, offering a promising foundation for future intelligent BMS. By pursuing the aforementioned research directions, the proposed PBPF can be further extended to robust battery state estimation under diverse operating conditions.

DATA AVAILABILITY

The datasets used in this study are publicly available benchmark battery datasets, including the LA92 driving cycle datasets at 40°C, 25°C, and 0°C, as well as the NASA B0005 cell dataset.

ACKNOWLEDGMENT

The authors acknowledge the facility support from the School of Applied Digital Technology, Mae Fah Luang University, Thailand.

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